



DeepReviewer 2.0: A Traceable Agentic System for Auditable Scientific Peer Review

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Code : ResearAI/DeepReviewer-v2

Video: Youtube: mMg5XzcaDCw

Project: <https://deepscientist.cc>

Abstract

Automated peer review is often framed as generating fluent critique, yet reviewers and area chairs need judgments they can *audit*: where a concern applies, what evidence supports it, and what concrete follow-up is required. DeepReviewer 2.0 is a process-controlled agentic review system built around an output contract: it produces a **traceable review package** with anchored annotations, localized evidence, and executable follow-up actions, and it exports only after meeting minimum traceability and coverage budgets. Concretely, it first builds a manuscript-only claim–evidence–risk ledger and verification agenda, then performs agenda-driven retrieval and writes anchored critiques under an export gate. On 134 ICLR 2025 submissions under three fixed protocols, an *untuned 196B* model running DeepReviewer 2.0 outperforms Gemini-3.1-Pro-preview, improving strict major-issue coverage (37.26% vs. 23.57%) and winning 71.63% of micro-averaged blind comparisons against a human review committee, while ranking first among automatic systems in our pool. We position DeepReviewer 2.0 as an assistive tool rather than a decision proxy, and note remaining gaps such as ethics-sensitive checks.

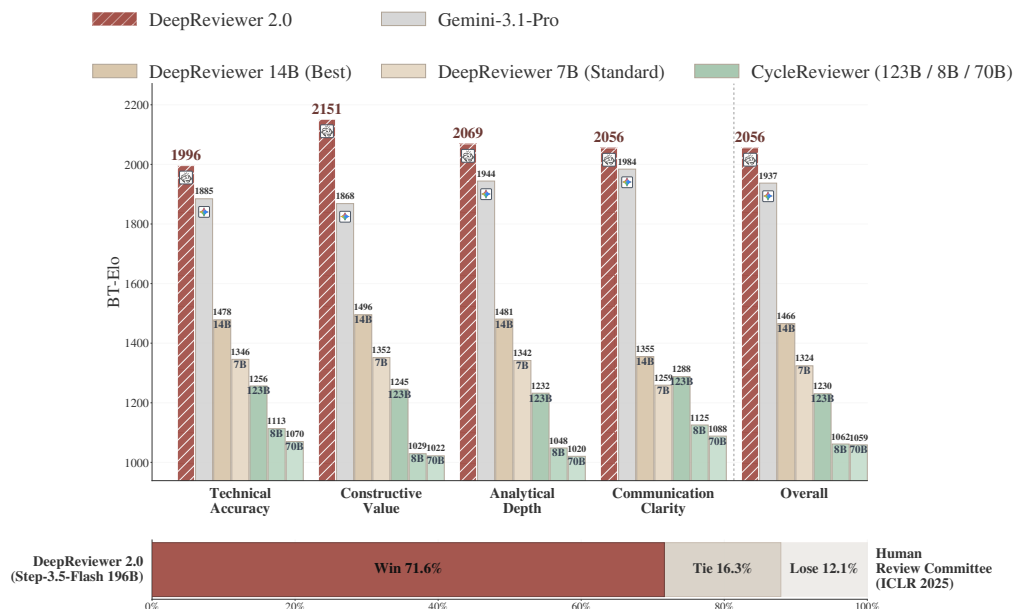


Figure 1: Dimension-wise Bradley–Terry Elo and blind preference against a human review committee.

1 Introduction

Peer review is not just a verdict; it is a justification that others must be able to trust. When a review claims “the experiments are insufficient” or “there is novelty overlap,” reviewers and area chairs still need to verify *where* this applies in the manuscript and *what* evidence supports the judgment before they can stand behind it. A usable review therefore has two properties: it is *auditable* (concerns are anchored to concrete locations and evidence) and *actionable* (it specifies concrete follow-up actions, such as missing baselines or required analyses). **Our work studies how to build automated reviewing around these properties rather than around fluent prose alone.**

Many automated reviewers are still built and evaluated mainly as one-shot generation or scoring systems. Even when the review sounds fluent, a common failure mode remains: the critique is hard to verify and hard to act on because it is not tied to the manuscript. If a system says “the experiments are insufficient” but cannot point to the relevant table, paragraph, or missing control, reviewers cannot safely endorse the judgment and authors do not know what to fix first. Recent LLM-based reviewers improve surface plausibility and guideline alignment, but most still do not *require* explicit evidence links as part of the review output (Yuan et al., 2021; Idahl & Ahmadi, 2024; Gao et al., 2025).

DeepReviewer 2.0 targets a different output. Compared to DeepReviewer 1.0 (Zhu et al., 2025), which already explored agentic retrieval for novelty checking, DeepReviewer 2.0 makes traceability a first-class requirement rather than a best-effort behavior. **We operationalize peer review as an agentic cognitive chain that produces a traceable review package.** The package includes a structured report, page/paragraph-anchored annotations, a prioritized repair plan, and an explicit novelty/value assessment. In Stage I, the system performs a manuscript-only pre-review and builds a claim–evidence–risk ledger. In Stage II, it turns that ledger into a verification agenda, checks novelty against closely comparable prior work under a matched-setting gate, and writes anchored annotations with concrete repair actions. Final synthesis is gated: the system does not export a package until minimum process requirements for retrieval, structure, and anchored feedback are satisfied (Section 3; Table 1).

This framing is intentionally conservative. **DeepReviewer 2.0 enforces process constraints, not semantic truth.** That distinction matters because automated reviewing can fail in ways that are not visible from prose quality alone, including hidden prompt injection in PDFs (Ye et al., 2024). It also matters at deployment time: pre-review tools can be miscalibrated or overused when treated as substitutes for peer review rather than assistive systems (Akella et al., 2025). By making evidence links explicit and conditioning export on minimum traceability requirements, we aim to improve what can be audited and executed, even when judgments remain fallible.

This leads to the central question of this paper: **Does making traceability and process constraints part of the review output contract improve diagnostic coverage and review usefulness under a fixed rubric?** We evaluate DeepReviewer 2.0 with three protocols aligned with this process view: strict issue coverage, anonymous ranking among automatic systems, and anonymous blind comparison against a human review committee under a shared rubric. Across these views, the results are consistent: DeepReviewer 2.0 ranks first among automatic systems in our pool, improves strict major-issue coverage over a Gemini-3.1-Pro baseline, and is preferred in blind comparison against the human committee.

Contributions.

- **Output contract.** We reframe automated reviewing around a traceable review-package interface (structured report, anchored annotations, repair plan, and novelty/value assessment), with traceability treated as a required output.
- **Process-controlled workflow.** We design a staged cognitive chain with an explicit claim–evidence–risk ledger, matched-setting retrieval for novelty checks, and an export gate that enforces minimum traceability requirements.
- **Three-protocol evaluation.** We evaluate DeepReviewer 2.0 with strict issue coverage, anonymous system ranking with Bradley–Terry strength estimation, and a blind comparison against a human review committee under the same rubric; under this fixed-rubric setup, DeepReviewer 2.0 improves strict major-issue coverage over a Gemini-3.1-Pro baseline, ranks first among automatic systems in our pool, and is preferred in blind comparison against the human committee (Tables 2 and 3).

System	Full review	Retrieval	Novelty check	Claim check	Page/para notes	Structured package	Traceable evidence
Yuan et al. (2021)	✓	×	×	×	×	○	×
OpenReviewer (Idahl & Ahmadi, 2024)	✓	×	×	×	×	✓	×
ReviewAgents (Gao et al., 2025)	✓	○	○	×	×	✓	○
AgentReview (Jin et al., 2024)	○	×	×	×	×	○	×
PaperReview.ai (Jiang & Ng, 2025)	✓	✓	✓	○	×	✓	×
DeepReviewer 1.0 (Zhu et al., 2025)	✓	✓	✓	○	×	○	○
DeepReviewer 2.0 (ours)	✓	✓	✓	✓	✓	✓	✓

Table 1: Functional comparison of representative automated reviewing systems (yes/partial/no). Here ✓ denotes an explicit, first-class capability in the system interface or output contract; ○ denotes partial/limited support; and × denotes absence. Assignments follow capabilities explicitly described by the original interfaces/outputs, not inferred latent behavior; we mark ✓ only when a capability is required by the interface/output contract, not merely achievable via free-form prompting. A fluent “full review” is not the same output target as an evidence-bound review package with traceable anchors.

Positioning. DeepReviewer 2.0 is intended as an assistive tool for reviewers and authors, not a replacement for peer review.

2 Related work

Automated scientific reviewing predates modern large language models, from early proposals for semi-automated review infrastructure to neural review generation (Alicea, 2013; Yuan et al., 2021). Recent LLM-based systems improve review fluency and often add rubrics, role decomposition, or stronger training/evaluation resources, as in OpenReviewer and ReviewAgents (Idahl & Ahmadi, 2024; Gao et al., 2025). Adjacent work also studies aggregation/meta-reviewing and audits failure modes of LLM-written reviews under realistic conditions (Hasan et al., 2024; Li et al., 2025). **Unlike this line of work, DeepReviewer 2.0 is not centered on fluent review generation alone; it targets a traceable review package as the primary output.**

A second thread treats reviewing as a process rather than a single generated artifact. AgentReview simulates reviewer–author–chair interactions to study how bias, expertise, and social dynamics affect outcomes (Jin et al., 2024). DeepReview decomposes reviewing into stages, including retrieval and verification for novelty checking (Zhu et al., 2025), while ReviewEval argues for multidimensional assessment of review quality (e.g., factuality and constructiveness) instead of a single score (Kirtani et al., 2025). **DeepReviewer 2.0 follows this process view, but makes anchored annotations and export-time traceability constraints mandatory parts of the system contract.**

Our work is also informed by evidence-grounded scholarly assistance, novelty assessment, and document grounding. Retrieval-augmented scholarly assistants and citation-quality studies show that usefulness depends on local evidence support (Weng et al., 2026), not only document-level relevance (Asai et al., 2024; Roy & Han, 2024). Novelty assessment has been operationalized as contribution-level comparison pipelines with explicit taxonomies (Wang et al., 2023; Zhao & Zhang, 2025), and paragraph/page-level localization relies on layout-aware document modeling and datasets (Xu et al., 2019; Pfizmann et al., 2022). **DeepReviewer 2.0 combines these ingredients for a different goal: an auditable peer-review artifact that jointly supports diagnosis, novelty checking, and actionable revision guidance.**

3 DeepReviewer 2.0

DeepReviewer 2.0 is built around a simple thesis: a usable peer review is not a monologue, but a sequence of constrained cognitive moves. A move may extract the paper’s core claims, check novelty against closely comparable prior work, or write a localized critique with an executable repair action. These moves are grounded by concrete requirements: judgments must be traceable to anchors in the paper, comparisons must use matched settings, and critiques must end with actionable revisions. The system therefore produces an auditable review package rather than a single free-form review text.

At a high level, DeepReviewer 2.0 parses the PDF into anchored semantic units, performs an independent pre-review to build a claim–evidence–risk ledger and an investigation agenda, then runs agenda-driven verification while writing anchored annotations. Final synthesis is gated before exporting the review package. Figure 2 summarizes the workflow.

Implementation details (MCP tool interfaces, the modified OpenAI Agent SDK agent loop, and PASA integration) are deferred to Appendix C.1 and Appendix C.2.

Stepping back, we define the task by three conditions that must hold together. **Soundness** asks whether major judgments close an internal evidence loop: each key critique must link to in-paper anchors (text spans, figures, tables, or equations), and missing anchors are treated as high-risk gaps. **Novelty/value** asks whether claimed contributions survive comparison to directly comparable prior work under matched settings; we record conservative novelty tags and default to “unclear” when evidence is insufficient. **Actionability** asks whether an author can act immediately: every anchored note ends with a concrete repair requirement (rewrite, clarification, ablation, added baseline, or reporting fix), not just a stylistic preference. DeepReviewer 2.0 enforces these as process constraints (schema, budgets, and traceability), not as semantic-truth guarantees.

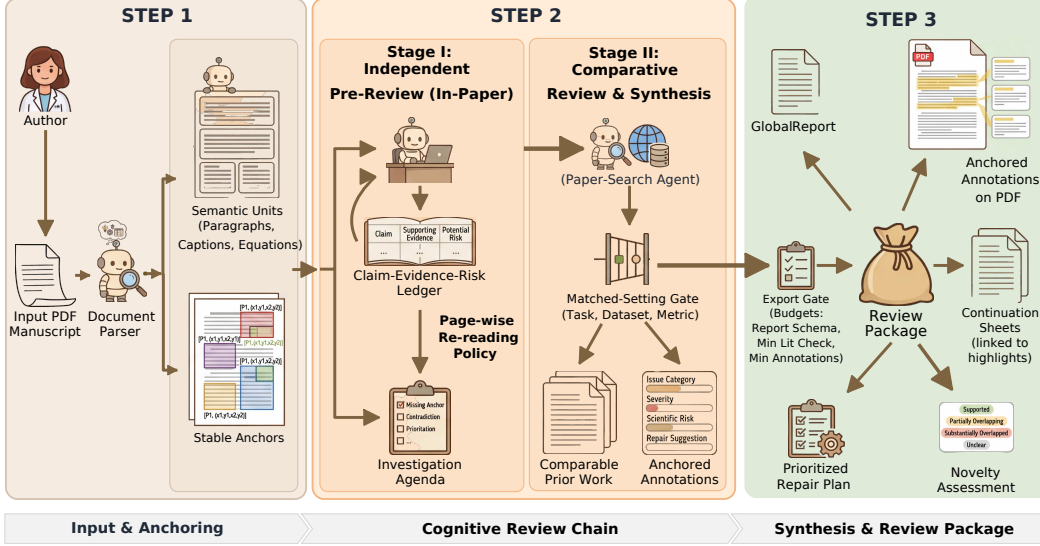


Figure 2: Overview of DeepReviewer 2.0 as an agentic cognitive chain. Stage I parses the PDF into anchored units and builds a claim–evidence–risk ledger plus an investigation agenda. Stage II performs agenda-driven verification (including matched-setting literature comparison) and writes anchored annotations with severity and concrete repair actions. Final synthesis is gated before exporting the review package.

3.1 Task formulation and notation

DeepReviewer 2.0 takes a paper in PDF form and turns it into a semantic representation that can be reasoned over while remaining traceable to the original layout. Concretely, we represent the paper as a sequence of semantic units (paragraphs, captions, equations with local context) paired with stable anchors back to the source. This view builds on progress in document layout understanding and scientific parsing (Xu et al., 2019; Pfizmann et al., 2022; Jimeno-Yepes et al., 2021); our contribution is to elevate anchors into a **required** part of the reviewing interface and downstream audit trail. We write the parsed paper as

$$X = \{(u_i, a_i)\}_{i=1}^N, \quad a_i = (p_i, b_i), \quad b_i \in \mathbb{R}_+^4, \quad (1)$$

where u_i is the i -th semantic unit and a_i stores its anchor: the page index p_i and a bounding box $b_i = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ under the parser’s page-local coordinate system. In our implementation, anchors are stored primarily as a page-and-line-span identifier (page index plus a contiguous line/paragraph range), optionally augmented with bounding boxes when layout coordinates are available for rendering. Anchors are defined with respect to the parser output rather than PDF viewer rendering, which enables paragraph-level feedback to be traced back to concrete regions of the source PDF. Appendix C details the anchored representation and overlay rendering.

The output is not a single free-form review paragraph. DeepReviewer 2.0 outputs a review package that bundles global judgment, local evidence, and actionable next steps:

$$Y = (\mathcal{R}, \mathcal{A}, \mathcal{P}, \mathcal{N}). \quad (2)$$

$\mathcal{R}$ is a structured final report (summary, strengths, weaknesses, overall judgment); $\mathcal{A}$ is a set of anchored annotations tied to specific paper locations; $\mathcal{P}$ is a prioritized repair plan; and $\mathcal{N}$ is a novelty/value assessment that makes comparisons explicit and records uncertainty when the evidence is thin. This output contract separates global synthesis from localized evidence, so readers can audit judgments without reconstructing the evidence trail manually.

Explicit evidence trail. DeepReviewer 2.0 treats the evidence trail as a first-class artifact, not as a latent reasoning trace. During the review, the system records typed links between (i) claims in the ledger, (ii) anchors in the paper, (iii) anchored annotations, and (iv) retrieved comparators used for novelty/value checks. We summarize this trail as a typed graph

$$G = (V, E), \quad V = V_{\text{claim}} \cup V_{\text{anchor}} \cup V_{\text{ann}} \cup V_{\text{prior}}, \quad (3)$$

where edges encode relations such as supported-by, contradicted-by, localized-to, and overlaps-with. A minimal traceability constraint is that every anchored annotation links to at least one paper anchor in V_{anchor} , so that a reader can jump from critique to evidence without reconstructing the chain. In practice, these links also support traceability checks before export (e.g., validating that major notes and judgments have at least one concrete anchor).

3.2 Stage I: Global diagnostic review

Stage I is an independent pre-review. DeepReviewer 2.0 first builds a paper map without leaning on external literature or other reviewers’ opinions. The output of this stage is a claim–evidence–risk ledger that makes the system commit to what it thinks the paper is claiming and what in-paper evidence is supposed to support those claims. Formally,

$$\mathcal{L}^{(0)} = f_{\text{meta}}(X) = \{(c_k, E_k^{(0)}, r_k^{(0)}, s_k)\}_{k=1}^K, \quad \mathcal{Q} = g(\mathcal{L}^{(0)}). \quad (4)$$

Each ledger entry contains a core claim c_k , its initial in-paper evidence set $E_k^{(0)}$ (anchors into X), an initial risk statement $r_k^{(0)}$ (what would break scientifically if the evidence is missing or misinterpreted), and a status label s_k . At this stage we use $s_k \in \{\text{confirmed}, \text{suspected}\}$ to separate defects that can be established from the manuscript alone versus issues that require verification. The stage also produces an investigation agenda $\mathcal{Q}$: a concrete list of questions to resolve in Stage II. This commits the system to explicit verification targets before final synthesis.

We keep the status criterion conservative. A defect is marked as “confirmed” when the paper provides a direct internal contradiction across anchors (e.g., inconsistent definitions or mismatched claims and tables), or when a core claim lacks any supporting anchor after a second pass. Otherwise it remains “suspected” and is elevated into the agenda. An example agenda item is: “Which closely comparable papers already report the same evaluation setting as the main claim, and does the contribution still hold under that comparison?” We prioritize agenda items by scientific risk: issues that can flip the validity of a central claim come first.

Page-wise re-reading policy. DeepReviewer 2.0 is not a single-pass generator. To reduce omission and to mitigate lost-in-the-middle failures, it iterates between selecting a paper span to (re)read and updating the ledger and annotations with the evidence found in that span:

$$\pi_t = \text{SelectSpan}(X, \mathcal{L}^{(t-1)}, \mathcal{Q}), \quad (\mathcal{L}^{(t)}, \mathcal{A}^{(t)}) = \text{Update}(\mathcal{L}^{(t-1)}, \mathcal{A}^{(t-1)}, X_{\pi_t}), \quad (5)$$

where π_t is a page-indexed location (page plus line/paragraph span). $\text{SelectSpan}(\cdot)$ prioritizes spans implicated by high-risk ledger items and agenda questions, while still advancing in page order to maintain coverage. This is the core operational bias: keep synthesis tethered to localized evidence, and revisit uncertain spans before finalizing the verdict. Here $\mathcal{A}^{(t)}$ denotes provisional anchored notes; final annotation writing and severity assignment are completed in Stage II.

3.3 Stage II: Verification-oriented annotation and synthesis

Stage II turns the agenda into evidence and action. For each agenda item $q_m \in \mathcal{Q}$ ($m = 1, \dots, M$), DeepReviewer 2.0 retrieves and reads a candidate set of papers and uses them to test novelty/value claims and to sanity-check methodological choices. We write this as

$$\mathcal{E}_m = \text{Retrieve}(q_m), \quad \hat{q}_m = \text{Verify}(q_m, X, \mathcal{E}_m). \quad (6)$$

In our implementation, $\text{Retrieve}(\cdot)$ is carried out by an academic paper-search agent (PASA) that expands an agenda question into targeted queries and consolidates evidence across sources (He et al., 2025). We define $M = |\mathcal{Q}|$.

Matched-setting comparability gate. We apply a matched-setting comparability gate before using prior work as novelty-overlap evidence. A paper is treated as directly comparable only if it matches the same task definition, the same dataset/benchmark, and the same primary evaluation metric. Other factors (e.g., model scale or compute budget) are used only as secondary ordering cues when such metadata is available, but they are not part of the gate. When comparability fails (e.g., task or metric mismatch), we may still record retrieved papers as background context, but we do not treat them as overlap evidence against a claim.

Conservative novelty tags. Given a comparable set, $\text{Verify}(\cdot)$ assigns one of four conservative novelty tags to each checked claim: **SUPPORTED** (no directly comparable prior work covers the core mechanism/claim under the matched setting), **PARTIALLY OVERLAPPING** (overlap exists but a clear residual novelty axis remains), **SUBSTANTIALLY OVERLAPPED** (directly comparable prior work already covers the core mechanism/claim under the matched setting), or **UNCLEAR**. The default is **UNCLEAR** when a claim cannot be mapped to a checkable mechanism or when evidence is insufficient or contradictory.

The goal is not generic similarity search, but matched-setting comparison under explicit comparability rules, aligned with evidence-grounded scholarly assistants and scientific fact-checking work (Asai et al., 2024; Wang et al., 2023; Vladika & Matthes, 2023). DeepReviewer 2.0 uses this stage to update the ledger and to decide what can be stated confidently versus what should be flagged for follow-up experiments or clearer positioning.

The main output artifact of Stage II is a set of anchored annotations. Each annotation is written as a small, structured unit so that it can be audited and acted on. We model an annotation as

$$a_j = (\ell_j, \tau_j, \sigma_j, \rho_j, \delta_j), \quad a_j \in \mathcal{A}. \quad (7)$$

ℓ_j is the location anchor (page/paragraph, potentially with a bounding box); τ_j is the issue category; σ_j is the severity (e.g., major/minor); ρ_j explains the scientific risk (why it matters, not just that it is “unclear”); and δ_j is the repair suggestion written so that an author can turn it into an explicit to-do. This bias toward actionability is consistent with annotation taxonomies and structured feedback settings that treat “what should I do next?” as the core value of critique (Choudhary et al., 2022; Tang et al., 2024). Operationally, we store ℓ_j as $(p_j, k_j^{\text{start}}, k_j^{\text{end}})$ (page plus line span) and only then derive a highlight rectangle for rendering. This ordering matters: page-and-span anchors remain stable under small extraction variations (e.g., missing boxes or minor line segmentation differences) produced by the PDF-to-markdown parser, while the layout rectangle is best-effort and can fall back to a conservative approximation when bounding boxes are unavailable (Appendix C).

Finally, anchored notes are exported as a readable, human-facing artifact by rendering the anchors back onto the source document. We overlay translucent highlights over the anchored spans and add right-margin callouts containing diagnosis and repair actions; when margin space is insufficient, the full text is moved to a continuation sheet linked to the source highlight. Appendix C.4 details the overlay rendering procedure.

3.4 Process constraints and auditability guarantees

DeepReviewer 2.0 is explicit about what it enforces and what it does not. The system is designed to enforce **process** properties: a structured report, minimum annotation coverage, and traceability links that let a reader check where a judgment came from. We can summarize the export gate for a review package as

$$Y = h\left(X, \mathcal{L}^{(*)}, \{\hat{q}_m\}_{m=1}^M, \mathcal{A}\right), \quad (8)$$

$$\mathbf{1}_{\text{ready}}(Y) = \mathbf{1}[\text{Schema}(\mathcal{R}) = 1] \cdot \mathbf{1}[n_{\text{search}} \geq \alpha] \cdot \mathbf{1}[n_{\text{intent}} \geq \beta] \cdot \mathbf{1}[|\mathcal{A}| \geq \gamma].$$

Here $\mathcal{L}^{(*)}$ denotes the ledger after Stage I/II updates, and $\text{Schema}(\mathcal{R})$ checks that the final report follows the required structure (e.g., summary, strengths, weaknesses, prioritized issues, and actionable suggestions). n_{search} and n_{intent} enforce a minimum literature-checking budget and distinct question

coverage for Stage II; and $|\mathcal{A}|$ enforces that the system produced a minimum amount of anchored, actionable feedback. In implementation, readiness checks also validate anchor integrity and per-annotation field completeness (risk explanation and repair action), not only aggregate counts. In the runs reported in this paper, we use $\alpha = 3$, $\beta = 3$, and $\gamma = 10$ as the enforced minimum budgets. In short, the goal is to make the output easier to audit, even if the underlying semantic judgments remain fallible.

The boundary is just as important. DeepReviewer 2.0 does not guarantee semantic truth, and it does not guarantee perfect novelty judgments. Automated reviewing is exposed to manipulation and systematic failure modes of language models, including prompt injection through hidden content in PDFs (Ye et al., 2024). It is also exposed to broader misalignment issues when a system designed for pre-review screening is treated as a substitute for peer review (Akella et al., 2025). The right way to read DeepReviewer 2.0, therefore, is as a process-controlled agentic review system: it can help reviewers and authors by making judgments more traceable and feedback more executable, but it should not be treated as an oracle.

4 Experiments

4.1 Data, protocol, and metrics

We evaluate on a sample of 134 ICLR 2025 submissions with three protocols aligned with our review-package contract: (i) strict issue coverage (diagnostic recall), (ii) anonymous chain ranking among automatic systems (end-to-end preference), and (iii) anonymous blind comparison against a human review committee under the same rubric. Strict coverage is computed on all 134 papers. Ranking metrics are computed on the subset where the Step-3 judge emits a valid comparison chain, and we use the same subset for both ranking protocols to avoid denominator drift (Appendix B.1).

Before ranking, we apply fixed pipeline validity checks (e.g., malformed review-package exports or unparseable judge chains are excluded). This keeps the protocol executable but can understate end-to-end brittleness in deployment (Section 6).

Strict issue coverage follows a conservative two-step rubric: we extract a canonical issue list from the human committee packet with semantic deduplication, severity (major/minor), and a single category label (Step-1), then score each system by strict binary coverage of each issue (Step-2), counting missing outputs as uncovered.

For anonymous ranking (Step-3), a fixed judge (gemini-3.1-pro-preview) produces comparison chains with ties on five dimensions (Kirtani et al., 2025; Chu et al., 2024; Zhang et al., 2025). We compute pairwise non-tie win fractions, uniform average win rate, tie-aware average rank, and Bradley–Terry Elo with 95% paper-level bootstrap confidence intervals (1000 resamples) (Selby, 2024). The human baseline is a committee-style packet (all reviews and discussion + rebuttal + meta-review/decision).

4.2 Results

We organize the analysis by four research questions, but the evidence should be read jointly. RQ1 measures strict diagnostic coverage against human-identified issues. RQ2 tests end-to-end preference among automatic systems under a fixed rubric. RQ3 places the same review package under blind comparison with a human review committee. RQ4 explains where gains come from (and where they do not) via category-level coverage.

RQ1: Does DeepReviewer 2.0 better cover human-identified issues under strict coverage, especially major issues? Yes. Table 2 reports strict issue coverage against the Step-1 human issue list. Under this conservative benchmark, the most important signal is major-issue coverage: DeepReviewer 2.0 achieves the best strict overall coverage (35.74%) and the best major-issue coverage (37.26%), improving by 13.69 points over a Gemini-3.1-Pro baseline (23.57%) and reducing the critical miss rate to 62.74%. The larger gains on major issues match the system design: the claim–evidence–risk ledger, risk-prioritized agenda, and page-wise re-reading bias the process toward decision-changing defects rather than maximizing the count of minor comments. At the same time, the absolute miss rate remains high under a strict binary matching policy, so we interpret

System	Overall (%)	Major (%)	Minor (%)	Critical miss (%)
DeepReviewer 2.0 (Step-3.5-Flash 196B)	35.74	37.26	30.98	62.74
CycleReviewer 123B	23.92	23.74	24.46	76.26
TheAIScientist Review (Gemini-3.1-Pro)	21.94	23.57	16.85	76.43
DeepReviewer 14B (Best)	21.42	23.40	15.22	76.60
AgentReview (Step-3.5-Flash 196B)	19.58	20.80	15.76	79.20
DeepReviewer 7B (Standard)	18.92	19.58	16.85	80.42
CycleReviewer 8B	11.17	12.48	7.07	87.52
CycleReviewer 70B	9.33	10.75	4.89	89.25

Table 2: Strict issue coverage against human-identified issues on the evaluation set. Missing system outputs are counted as uncovered. Higher is better for Overall/Major/Minor; lower is better for Critical miss rate ($1 - \text{Major}$).

RQ1 as improved diagnostic recall under a conservative protocol, not as evidence of human-level completeness.

RQ2: Under anonymous ranking among automatic systems, is DeepReviewer 2.0 consistently preferred? Yes. We summarize anonymous ranking among automatic systems with three complementary views (pairwise breadth, aggregate robustness, and metric agreement), each computed from the same Step-3 judge chains.

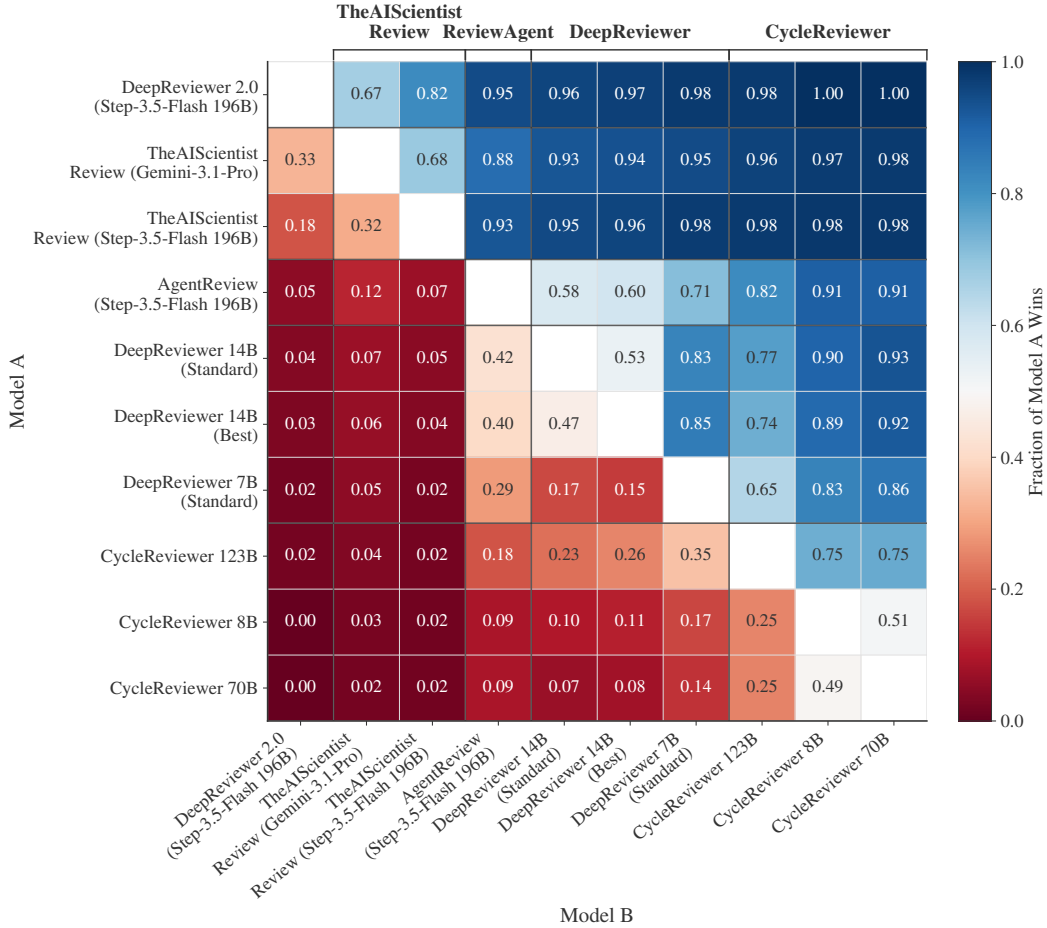


Figure 3: Pairwise non-tie win-fraction heatmap among automatic systems.

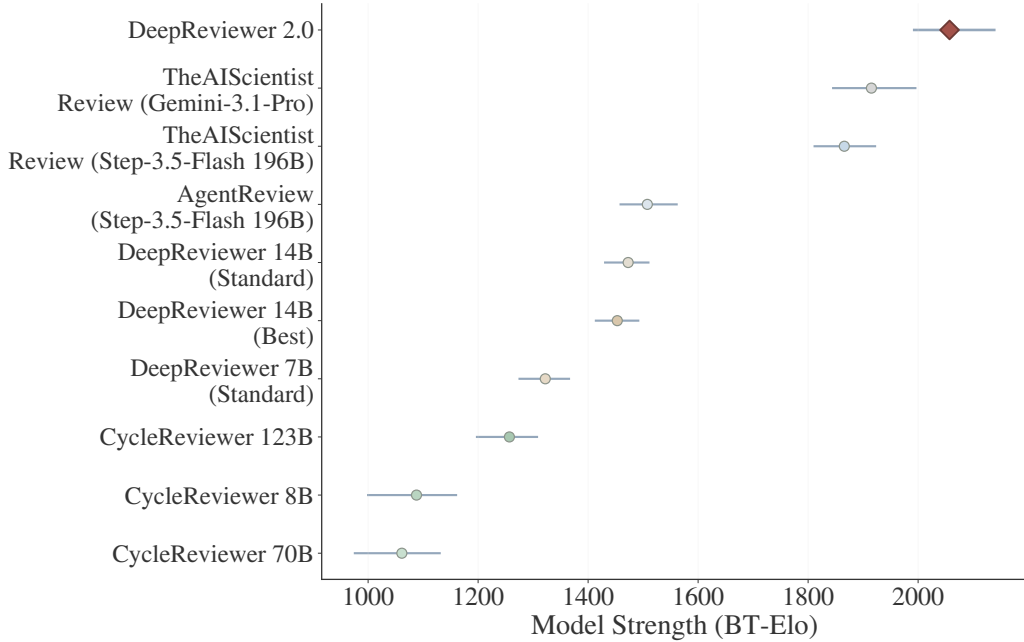


Figure 4: Bradley-Terry model strength estimation (Elo) with 95% bootstrap confidence intervals (1000 paper-level resamples).

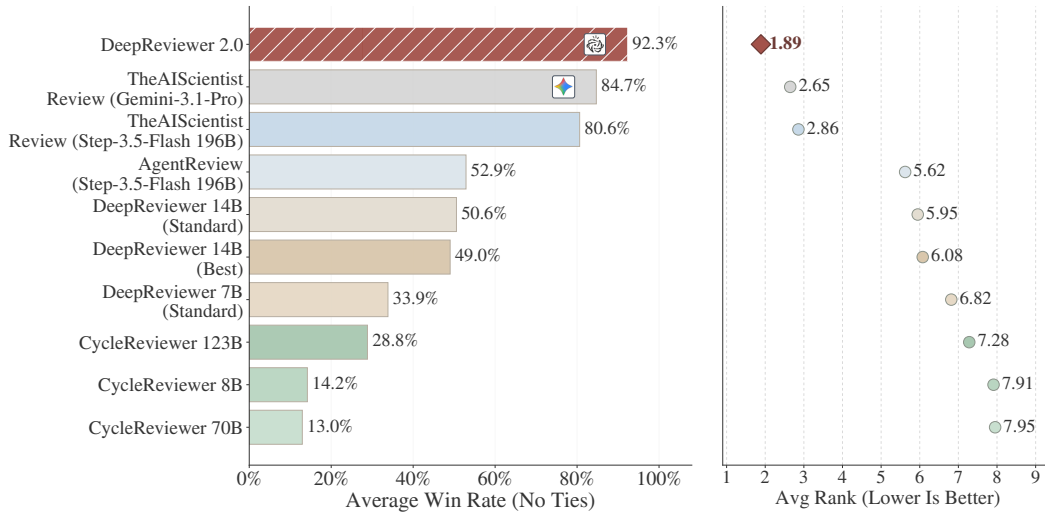


Figure 5: Two-panel summary of anonymous ranking among automatic systems: uniform average non-tie win rate (left) and average rank (right; lower is better).

Pairwise breadth (Figure 3). DeepReviewer 2.0 wins broadly across opponents; importantly, its hardest opponent in this pool is the Gemini-3.1-Pro baseline, against which it still wins 66.86% of non-tied comparisons. This argues against a narrow gain that only holds against weak baselines.

Aggregate robustness (Figure 4). Under an order-invariant strength model, DeepReviewer 2.0 remains the top-ranked system with 2057.17 Elo (95% CI: [1990.55, 2140.81]), ahead of the strongest Gemini-3.1-Pro baseline (1915.11 Elo), with a 142.07 Elo gap. This supports the claim that preference is stable under aggregation and not an artifact of chain ordering.

Metric agreement (Figure 5). DeepReviewer 2.0 leads under both uniform average non-tie win rate (92.34%) and tie-aware average rank (1.889; lower is better), reducing the chance that preference

is a single-metric artifact. Taken together, Figures 3–5 support a stronger conclusion than “wins on average”: under a fixed judge and tie rule, DeepReviewer 2.0 is preferred broadly across opponents and remains first under multiple aggregations. We still treat this as judge-conditional evidence of end-to-end usefulness, not as a direct measure of factual correctness.

RQ3: In anonymous blind comparison, how does DeepReviewer 2.0 compare to a human review committee? We compare DeepReviewer 2.0 against a human review committee using the same anonymous ranking protocol, and report win/tie/lose percentages in Table 3. On the micro-average across all dimensions, DeepReviewer 2.0 is preferred in 71.63% of comparisons (ties 16.28%, losses 12.09%). The dimension breakdown is more informative than the micro-average alone. The strongest advantages are in Constructive Value (84.50% win) and Communication Clarity (86.05% win), consistent with the design goal of producing localized, actionable feedback with an executable repair path. By contrast, margins are smaller in Technical Accuracy (59.69%) and Analytical Depth (58.14%), suggesting that process constraints improve evidence organization and revision guidance more than they solve all technical reasoning errors. We therefore interpret RQ3 as a preference signal under this rubric and judge setting, not as a claim of general human replacement.

Table 3: Anonymous blind preference comparison between DeepReviewer 2.0 and a human review committee (win/tie/lose percentages).

Dimension	DeepReviewer 2.0 win (%)	Tie (%)	Human win (%)
All Dimensions (micro)	71.63	16.28	12.09
Technical Accuracy	59.69	24.81	15.50
Constructive Value	84.50	5.43	10.08
Analytical Depth	58.14	24.03	17.83
Communication Clarity	86.05	9.30	4.65
Overall Judgment	69.77	17.83	12.40

RQ4: What does the category-level coverage profile reveal about strengths and gaps? Figure 6 breaks strict coverage down into eight issue categories and shows that DeepReviewer 2.0 is not uniformly best in every category. It leads in 4/8 categories (Novelty, Clarity, Presentation, and Other), while other systems remain stronger in Soundness/Experiments and in ethics-sensitive checks. This profile also explains why a system can lead overall without being best everywhere: the canonical issue list is highly imbalanced (e.g., Experiments has 225 issues and Soundness has 138 issues, while Ethics has only 6 issues). DeepReviewer 2.0’s overall advantage therefore comes from concentrated strengths in high-frequency categories together with competitive (though not always best) coverage elsewhere.

The most salient blind spot is Ethics: DeepReviewer 2.0 scores 0.00% strict coverage there under this protocol, while another system reaches 50.00%. Even though the category is small in the current benchmark, it matters disproportionately for deployment risk. We therefore read Figure 6 as both an explanation of where the current gains come from and a concrete target for improvement: without explicit objectives and dedicated evaluation pressure, ethics-sensitive critique is easy to underweight.

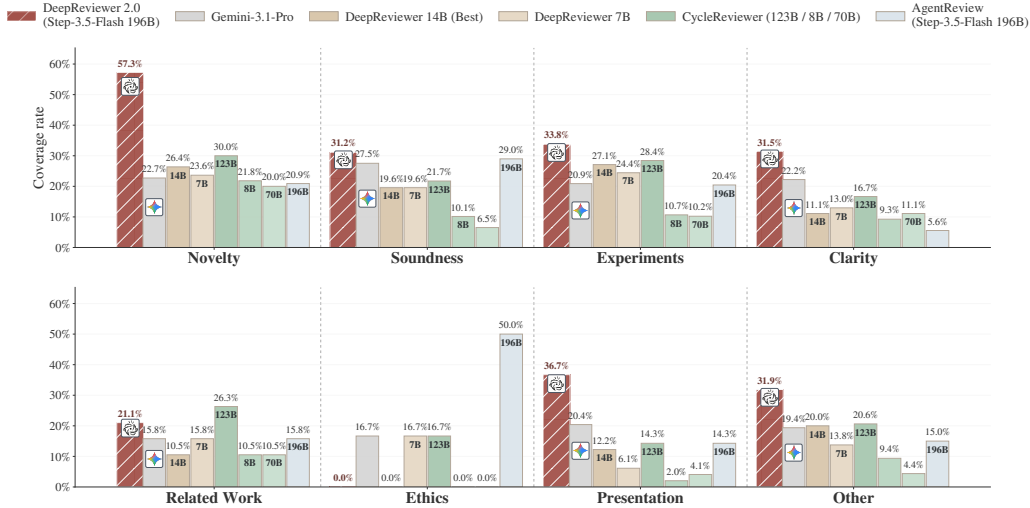


Figure 6: Category-level strict issue coverage across eight issue categories.

5 Conclusion

DeepReviewer 2.0 argues that automated reviewing should be judged less by how persuasive it sounds and more by whether its judgments are checkable. We operationalize this stance with a **review-package** interface and a staged cognitive chain that makes traceability budgets and gated synthesis part of the output contract. Across strict issue coverage, anonymous system ranking, and a blind comparison against a human review committee, the resulting packages are preferred under our protocol. At the same time, strict coverage remains far from complete and ethics coverage can fail sharply; DeepReviewer 2.0 is therefore best read as an **assistive reviewer** whose value depends on how it is integrated with human oversight and venue-specific checklists.

6 Limitations and Ethical Risk Considerations

DeepReviewer 2.0 is constrained by design, and some constraints are the point of the paper. The system provides **process-level** guarantees (structured outputs and traceability links), not semantic truth guarantees: a review package can be coherent and still be wrong about novelty, factual claims, or scientific priorities. This matters because strict issue coverage remains incomplete—even under our protocol, major issues can be missed—so DeepReviewer 2.0 should be used as an assistive tool to augment human reviewers rather than to replace them (Akella et al., 2025).

Our evaluation protocols also define what the results do and do not measure. Strict issue coverage is computed against a canonical issue list extracted from human reviews/discussion plus the meta-review/decision, which rewards matching what humans already articulated but does not credit newly discovered issues. The anonymous ranking protocols inherit the judge model, rubric, and tie rule; we therefore report multiple complementary aggregates (pairwise win fractions, tie-aware average rank, and an order-invariant Bradley–Terry strength estimate with bootstrap intervals), but different judges could shift preferences, especially on technical accuracy (Kirtani et al., 2025; Chu et al., 2024). Because our judge is Gemini-family while the comparison pool includes a Gemini-3.1-Pro baseline, preferences may also reflect stylistic affinity in addition to substantive quality.

In addition, our quantitative results are conditional on a fixed evaluation pool (Appendix B.1) that excludes malformed exports and unparseable ranking outputs. This simplifies protocol execution and metric computation, but it can understate end-to-end reliability failures (e.g., tool failures, parser brittleness, or non-executable packages) that matter in unconstrained deployment.

Ethical and deployment risks. Automated reviews can be misused as gatekeeping signals or decision proxies, especially when outputs sound confident but remain fallible. The system is also exposed to manipulation, including hidden-content prompt injection in PDFs (Ye et al., 2024), and to privacy and data-handling concerns when manuscripts are routed through toolchains for search, retrieval,

and rendering. Our category analysis further reveals a sharp gap on Ethics (Figure 6), suggesting that ethics-sensitive critique is easy to underweight when technical content dominates unless it is explicitly pressured by objectives and evaluation. A conservative stance is to treat high-stakes or normatively loaded domains (e.g., medical, legal, dual-use AI) as out of scope unless dedicated safeguards are added on top of technical critique, such as PDF sanitization, explicit ethics checklists, and human oversight with domain expertise.

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A Evaluation instruction templates

This appendix summarizes the instruction templates used by the automated judges in Step-2 (strict issue coverage) and Step-3 (anonymous ranking). We state them as evaluation contracts (inputs, decision rules, outputs). The intent is not to imply that results depend on phrasing tricks, but to make the protocol concrete enough to reproduce.

A.1 Strict issue-coverage evaluator (Step-2)

Inputs. The evaluator receives (i) a canonical issue list extracted from human reviews and (ii) one system-generated review text for the same paper.

Decision rule. Each issue is judged independently with a strict binary criterion. The evaluator assigns label = 1 only when the system review clearly covers the same issue with concrete semantic overlap; label = 0 is used when the issue is missing, too vague, or only weakly related. External knowledge is disallowed: decisions must rely only on the issue description and the system review text.

Output contract. The evaluator returns a single structured record with one entry per issue (issue id, binary label, short reason, optional evidence span from the system review), plus a brief overall assessment. Hard constraints include: every issue id appears exactly once; labels are in $\{0, 1\}$; partially related mentions do not count as coverage unless the core concern is addressed.

A.2 Anonymous ranking judge (Step-3)

Inputs and anonymity. The judge compares multiple candidate reviews for the same paper, treating every candidate as an automated-system output. Candidate identities are anonymized, and the judge is instructed not to use prior assumptions about model family, style, or authorship.

Weakness-driven comparison. The judge reads the provided paper text and candidate reviews, then conducts a weakness-by-weakness analysis: it first builds a canonical weakness inventory from the paper and all candidates (splitting fundamental root-cause issues from secondary issues), then maps each candidate’s weakness claims to that inventory and assesses correctness, materiality, evidence quality, repairability, and actionability. The judge focuses on substantive review quality and is instructed to ignore minor formatting or surface-level artifacts unless they change meaning.

Ranking chains and tie rule. The judge outputs five aspect-specific ranking chains (Technical Accuracy, Constructive Value, Analytical Depth, Communication Clarity, Overall Judgment) using only “>” and “=”. Ties are governed by an explicit conversion rule: the judge uses internal integer scores in $[1, 10]$; score gaps ≤ 1 must be rendered as ties “=”, and only gaps ≥ 2 may produce strict “>” relations.

B Evaluation metrics and computation details

This appendix spells out the metric definitions and computation rules used in the strict-coverage (Step-2) and anonymous-ranking (Step-3) protocols.

B.1 Notation and dataset filtering

We evaluate on a sample of 134 ICLR 2025 submissions. Let $\mathcal{P}$ denote the set of evaluated papers, so $|\mathcal{P}| = 134$. The ranking-based protocols use a fixed evaluation pool $\mathcal{P}_{\text{rank}}$ consisting of the subset of $\mathcal{P}$ whose Step-3 judge outputs are parseable, and we use the same $\mathcal{P}_{\text{rank}}$ pool for anonymous ranking among systems and for the blind comparison against a human review committee.

For each paper $p \in \mathcal{P}$, Step-1 yields a canonical issue list $\mathcal{I}_p$, where each issue has (i) a severity tag (major/minor) and (ii) a single category from an eight-way taxonomy.

B.2 Step-2 coverage metrics (strict policy)

For a system s , paper p , and issue $i \in \mathcal{I}_p$, Step-2 produces a label

$$y_{s,p,i} \in \{0, 1, \emptyset\}, \quad (9)$$

where $\emptyset$ denotes missing or unparseable output. Under the strict policy, $\emptyset$ is treated as uncovered.

Overall coverage.

$$\text{OverallCoverage}_s = \frac{\sum_{p \in \mathcal{P}} \sum_{i \in \mathcal{I}_p} \mathbf{1}[y_{s,p,i} = 1]}{\sum_{p \in \mathcal{P}} |\mathcal{I}_p|}. \quad (10)$$

Missing/unparseable output ($\emptyset$) is counted as uncovered under the strict policy.

Severity-split coverage and critical miss rate. Let $\mathcal{I}_p^{\text{major}}$ and $\mathcal{I}_p^{\text{minor}}$ be the major and minor issue subsets for paper p .

$$\text{MajorCoverage}_s = \frac{\sum_p \sum_{i \in \mathcal{I}_p^{\text{major}}} \mathbf{1}[y_{s,p,i} = 1]}{\sum_p |\mathcal{I}_p^{\text{major}}|}, \quad (11)$$

$$\text{MinorCoverage}_s = \frac{\sum_p \sum_{i \in \mathcal{I}_p^{\text{minor}}} \mathbf{1}[y_{s,p,i} = 1]}{\sum_p |\mathcal{I}_p^{\text{minor}}|}, \quad (12)$$

$$\text{CriticalMissRate}_s = 1 - \text{MajorCoverage}_s. \quad (13)$$

Category coverage. For category c , let $\mathcal{I}_{p,c}$ denote the issues in paper p that fall into category c . We compute

$$\text{CatCoverage}_{s,c} = \frac{\sum_p \sum_{i \in \mathcal{I}_{p,c}} \mathbf{1}[y_{s,p,i} = 1]}{\sum_p |\mathcal{I}_{p,c}|}. \quad (14)$$

B.3 Step-3 ranking metrics

For each paper p and aspect a , the judge returns a total preorder over systems, represented as a chain with operators “ $>$ ” and “ $=$ ”. A tie tier is a maximal contiguous group linked by “ $=$ ”.

Average rank with tie tiers. Suppose system s appears in a tie tier occupying positions $r, \dots, r + k - 1$ in the chain (lower is better). We assign the tier-average rank

$$\text{Rank}_{p,a}(s) = \frac{r + (r + k - 1)}{2}, \quad (15)$$

and define AvgRank_s as the mean of $\text{Rank}_{p,a}(s)$ over all valid (p, a) instances.

Pairwise non-tie win fraction. Let $W_{x,y}$ be the count of strict wins of system x over y aggregated over papers and aspects, counting only “ $>$ ” relations and excluding ties. The pairwise non-tie win fraction is

$$\text{WinFrac}_{x,y} = \frac{W_{x,y}}{W_{x,y} + W_{y,x}}. \quad (16)$$

We exclude ties here because, under the judge protocol, ties are induced by score gaps within the judge’s 1–10 scoring resolution ($\text{gap} \leq 1$), and treating them as half-wins can blur decisive preference signals.

Uniform average win rate. Let $\mathcal{B}_x$ be the set of opponents of x in the compared pool. The uniform average win rate is

$$\text{AvgWinRate}_x = \frac{1}{|\mathcal{B}_x|} \sum_{y \in \mathcal{B}_x} \text{WinFrac}_{x,y}. \quad (17)$$

B.4 Bradley–Terry Elo estimation and bootstrap confidence intervals

We do not use an online Elo update. Instead, we treat non-tie comparisons as outcomes of a Bradley–Terry model with latent abilities $\{\theta_s\}_s$:

$$\Pr(x \succ y) = \frac{\theta_x}{\theta_x + \theta_y}. \quad (18)$$

Let $W_{x,y}$ be the non-tie win counts as above, and let $N_{x,y} = W_{x,y} + W_{y,x}$. We estimate θ by maximum likelihood and solve it using a standard minorization–maximization (MM) iteration:

$$\theta_x^{(t+1)} = \frac{\sum_{y \neq x} W_{x,y}}{\sum_{y \neq x} \frac{N_{x,y}}{\theta_x^{(t)} + \theta_y^{(t)}}}. \quad (19)$$

To report results on the familiar Elo scale, we map

$$\text{Elo}_s = 1500 + 400 \log_{10}(\theta_s), \quad (20)$$

where we normalize θ by a constant factor so that the geometric mean ability is 1 (this fixes the BT scale ambiguity without changing relative strengths).

Bootstrap confidence intervals. We compute 95% confidence intervals by paper-level bootstrap with 1000 resamples (sampling papers with replacement). For each resample, we rebuild the win matrix, re-fit the BT model, re-compute Elo, and report the percentile interval [2.5%, 97.5%].

B.5 DeepReviewer 2.0 vs human committee percentages

For each aspect a , we compare DeepReviewer 2.0 against the human review committee by their ranks in the judge chain on each paper. Let w_a , t_a , and ℓ_a be the counts over papers where $\text{Rank}(\text{DeepReviewer 2.0}) < \text{Rank}(\text{Human})$, equal, or greater, respectively. The reported percentages are normalized per row:

$$\text{WinPct}_a = \frac{w_a}{w_a + t_a + \ell_a}, \quad \text{TiePct}_a = \frac{t_a}{w_a + t_a + \ell_a}, \quad \text{LosePct}_a = \frac{\ell_a}{w_a + t_a + \ell_a}. \quad (21)$$

Micro-average across dimensions. The “All Dimensions (micro)” row aggregates counts across aspects:

$$\text{WinPct}_{\text{micro}} = \frac{\sum_a w_a}{\sum_a (w_a + t_a + \ell_a)}, \quad (22)$$

$$\text{TiePct}_{\text{micro}} = \frac{\sum_a t_a}{\sum_a (w_a + t_a + \ell_a)}, \quad (23)$$

$$\text{LosePct}_{\text{micro}} = \frac{\sum_a \ell_a}{\sum_a (w_a + t_a + \ell_a)}. \quad (24)$$

C Document parsing and anchored annotation

This appendix clarifies how DeepReviewer 2.0 turns a PDF into a traceable text representation and how paragraph-level feedback is rendered as anchored annotations. We focus on the representation contract: what information is carried forward so that every critique can be traced back to a concrete page region, even when the input PDF is noisy.

C.1 From PDF to page-indexed semantic text

DeepReviewer 2.0 normalizes each input PDF into (i) a page-indexed semantic text stream and (ii) a layout trace produced by the PDF parser. In our implementation, we use MinerU (v4) to produce markdown and a `content_list` of extracted blocks; each block records its page index, text, type (e.g., header/text/equation/table/image), and (when available) a bounding box. We then build a page index by ordering the extracted texts per page:

$$\mathcal{J}(p) = [t_{p,1}, t_{p,2}, \dots, t_{p,L_p}], \quad (25)$$

where L_p is the number of extracted lines on page p . When layout coordinates are available, each line $t_{p,k}$ additionally carries a bounding box $\text{BBox}(p, k) = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ together with per-page reference scales (W_p, H_p) from the parser output. This dual view is deliberate: the text stream supports reasoning and comparison, while the optional boxes support rendering highlights; when boxes are missing or noisy, traceability still holds through stable page-and-line anchors. Coordinates follow a top-left origin convention with x increasing rightward and y increasing downward.

In practice, the parser returns bounding boxes in a page-local coordinate system with an implicit reference scale. We treat each box as $b_{p,k} = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ together with a per-page reference (W_p, H_p) (computed from the maximum observed extents on that page), so that we can normalize by (W_p, H_p) and map highlights back onto the source PDF consistently during export (Section C.3). Figure 7 visualizes this parsing contract on a real paper: each semi-transparent rectangle is a MinerU-returned block box on the first four pages of arXiv:2509.26603.

C.2 Implementation pseudocode (MCP tool loop)

Figure 8 summarizes the end-to-end pipeline as executed in our backend. The key point is that “real-world” interactions are not implicit: the agent reads pages and writes annotations through MCP tools, and the export gate is checked over explicit counters and artifacts (Section 3).

C.3 Annotation objects and anchor resolution

Each annotation couples a location anchor with a diagnosis and a repair action. In the main paper we write an annotation as $a_j = (\ell_j, \tau_j, \sigma_j, \rho_j, \delta_j)$ (Equation 7). Operationally, the location ℓ_j is a page and line span:

$$\ell_j = (p_j, k_j^{\text{start}}, k_j^{\text{end}}), \quad 1 \leq k_j^{\text{start}} \leq k_j^{\text{end}} \leq L_{p_j}. \quad (26)$$

The annotation also carries a short summary and a longer comment text used for the callout body. For example, an annotation may target page 2, lines 14–22, be marked as a major issue, summarize an overstated novelty claim, explain the scientific risk, and then specify a concrete revision requirement (often including a proposed rewrite).

To render a highlight overlay, we convert $(p_j, k_j^{\text{start}}, k_j^{\text{end}})$ into a set of rectangles on the source PDF page. When parser bounding boxes are available, we take the union of the boxes covered by the line range:

$$\mathcal{R}_j = \bigcup_{k=k_j^{\text{start}}}^{k_j^{\text{end}}} \text{BBox}(p_j, k). \quad (27)$$

When bounding boxes are unavailable, we fall back to a line-ratio approximation that allocates a vertical span proportional to the line indices:

$$y_1 = 100 \cdot \frac{k_j^{\text{start}} - 1}{L_{p_j}}, \quad y_2 = 100 \cdot \frac{k_j^{\text{end}}}{L_{p_j}}, \quad (28)$$

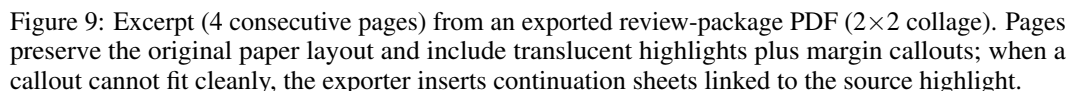
and use a conservative fixed horizontal range (e.g., $x \in [8, 92]$ in a normalized 0–100 coordinate system) so that the highlight remains visible even under extraction noise.

C.4 Overlay rendering and continuation pages

DeepReviewer 2.0 exports anchored feedback as a combination of (i) translucent highlights over the source page regions and (ii) right-margin callouts that carry the critique text. Each callout receives a stable marker (page index plus per-page annotation index), and the renderer routes a short connector line from the highlighted region to the corresponding callout while prioritizing non-overlap among callout boxes.

Long comments are common in reviewer-style writing. When a callout cannot fit in the right margin without overlap, DeepReviewer 2.0 switches to a continuation mode: the source page keeps a compact index card indicating that the full text is moved, and the complete comment is placed on a continuation sheet linked back to the source highlight. This yields an annotated PDF where the source page remains readable while the critique retains enough detail to be executable.

We include a representative case study to make the review package interface concrete. The example comes from a generated review of an arXiv preprint (arXiv:2509.26603). We reproduce (i) two four-page excerpts from the exported PDF that illustrate anchored highlights and continuation sheets, and (ii) the complete structured review report.



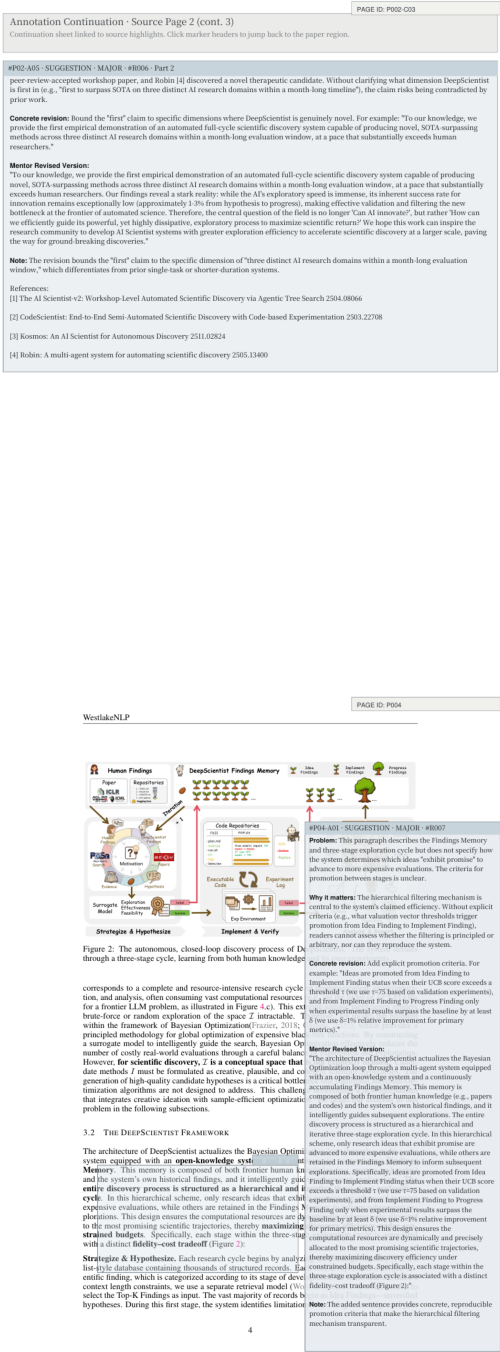


Figure 10: Additional excerpt (4 consecutive pages) from the same exported review-package PDF (2x2 collage). This span illustrates how anchored critiques can extend across sections while keeping the source pages readable and the repair actions explicit.

The full report illustrates the global layer: a structured summary followed by strengths, weaknesses, and a decision recommendation. This is where DeepReviewer 2.0 tries to stay honest about what it does and does not know. It is tempting for automated systems to write confident prose; the report format forces a more checkable decomposition (e.g., explicitly separated novelty concerns, methodological transparency concerns, and statistical reporting concerns). In this case, the global judgment depends heavily on whether the paper’s “first” claims survive direct comparison to closely related autonomous-scientist systems under matched settings. Instead of leaving novelty as an

impressionistic statement, the report records concrete comparators and qualifies verdicts when direct comparability is unclear.

Figures 9 and 10 illustrate the local layer: each callout follows a repeatable structure that mirrors our cognitive-chain design. A paragraph role diagnosis explains what that passage is supposed to do in the paper; a concrete defect states what is wrong; a risk paragraph explains why it matters scientifically; and a revision requirement gives a specific action the authors can execute, often accompanied by a proposed rewrite. The continuation sheets make this format practical: the source page stays readable, but the critique retains enough detail to be actionable. This is the core interface claim of DeepReviewer 2.0: we are not trying to “sound like a reviewer”; we are trying to leave behind a trail that another human can check, disagree with, and still use to revise the paper.

WeLaiNLP

DEEPSKIENTIST: ADVANCING FRONTIER-PUSHING SCIENTIFIC FINDINGS PROGRESSIVELY

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ABSTRACT

While previous AI Scientist systems can generate novel findings, they often lack the focus to produce scientifically valuable contributions that address pressing human-defined challenges. We introduce DeepScientist, a system designed to overcome this by conducting goal-oriented, fully autonomous scientific discovery over month-long timelines. It formalizes discovery as a Bayesian Optimization problem, operationalized through a hierarchical evaluation process consisting of ‘hypothesize, verify, and analyze’. Leveraging a cumulative Findings Memory, this loop intelligently balances the exploration of novel hypotheses with exploitation, selectively promoting the most promising findings to higher-fidelity levels of validation. Consuming over 20,000 LLM tokens, the system generated about 5,000 unique scientific ideas and experimentally validated approximately 1,100 of them, ultimately surpassing human-designed state-of-the-art (SOTA) methods on three frontier AI tasks by 183.7%, 1.9%, and 7.9%. This work provides the first large-scale evidence of an AI achieving discoveries that progressively surpass human SOTA on scientific tasks, producing valuable findings that genuinely push the frontier of scientific discovery.

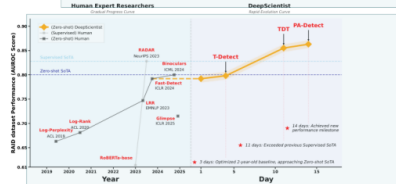


Figure 1: Comparison of research progress timelines for AI text detection on the RAID (Weng et al., 2025). The left panel shows that DeepScientist achieves progress in two weeks that is comparable to three years of human research (Sun et al., 2024; Sun et al., 2023) (left panel). All zero-shot methods, including the system-generated T-Detect, TDT, and PA-Detect, uniformly outperform the baseline (Human Expert Research). Additionally, all methods produced by DeepScientist demonstrate higher throughput than the previous SOTA method, Binoculars (Huan et al., 2024).

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2 RELATED WORK

Replication and Optimization. A significant body of research focuses on engineering tasks that operate within established scientific frameworks. This includes replication-oriented works like PaperPilot (Vassire et al., 2023) and PaperAgent (Mao et al., 2023), which aim to reproduce existing papers. Other works, such as Agent Laboratory (Chen et al., 2023) and MLL-Bench (Chen et al., 2024), tackle early-stage machine learning engineering problems. Similarly, systems like AlphaFlow (Weng et al., 2023) and AlphaProbe (Doroshenko et al., 2023) use massive trial-and-error with known engineering methods to improve the performance of codebases. The common goal of these efforts is engineering-driven optimization within an established scientific paradigm, enhancing existing systems without questioning their foundational assumptions. DeepScientist, in contrast, pursues scientific discovery by targeting the core limitations of the SOTA itself. Its objective is not to refine the current state-of-the-art, but to establish a new one by introducing fundamentally different methodologies.

Semi-Automated Scientific Assistance. The path toward automating scientific discovery begins not with replacing the scientist, but with assisting them, leading to the development of a paradigm of specialized AI tools for individual research tasks. Systems like CycleResearcher (Weng et al., 2025) handle writing. DeepReviewer (Zhu et al., 2025a) manages reviewing, and co-scientists (Goffredo et al., 2025; Pissarello et al., 2025; Svensson et al., 2025; Bick et al., 2025) aid in hypothesis generation. These powerful tools address only isolated fragments of the scientific process, leaving the crucial loop of learning from failure and exploration to humans. In contrast, DeepScientist is an autonomous agent of inquiry, managing the entire end-to-end research cycle and closing the loop by learning from its own experiments and self-directing its research path.

Automated Scientific Discovery. Building on the capabilities of specialized assistants, a line of research pursues full, end-to-end research automation (Yang et al., 2023; Xu et al., 2025a). Pioneering efforts, such as the AI Scientist systems (Lin et al., 2024; Yamada et al., 2023) and subsequent work (Huang, 2023; Jahan et al., 2023), successfully demonstrate that an AI system could manage the full research cycle and produce novel findings. However, their primary limitation often lies in their exploratory strategy, which lacks a specific scientific goal rooted in a field’s grand challenges, resulting in undirected discoveries that may be perceived as lacking genuine scientific value. Instead, DeepScientist is the first automated scientific discovery system that leverages a closed-loop, iterative process to discover methods, surpassing the human state-of-the-art. The exploration of DeepScientist is goal-oriented and insight-driven, beginning by identifying a recognized limitation in the human SOTA and then using failure attribution to ensure discoveries are both novel and scientifically meaningful.

3 DEEPSKIENTIST: A PROGRESSIVE SYSTEM FOR DISCOVERING SOTA-SURPASSING FINDINGS

3.1 MODELING SCIENTIFIC DISCOVERY AS AN OPTIMIZATION PROBLEM

The fundamental goal of automated scientific discovery is to autonomously identify novel methods that yield significant advancements in a given scientific domain. This process can be formally conceptualized as a search for an optimal solution within a vast and unstructured space of possibilities. Let the space of all possible candidate research methods be denoted by Z . Each individual method $f \in Z$, such as a novel algorithm or a new model architecture, possesses an intrinsic scientific value. This value is determined by a latent, black-box true value function, $f^* : Z \rightarrow \mathbb{R}$, which maps a method to its ultimate empirical impact. The objective of scientific discovery is therefore to find the optimal method f^* that maximizes this function:

$$f^* = \arg \max_{f \in Z} f^*(f) \quad (1)$$

Unlike previously studied tasks such as early-stage machine learning (Schneidman et al., 2025a), algorithmic design (Novikov et al., 2025; Lange et al., 2025), or scientific software development (Vignuzzi et al., 2025), defining characteristic of frontier scientific discovery is that each exploratory step demands immense computational and intellectual resources, making the evaluation of the true scientific value function, $f^*(\cdot)$, prohibitively costly. Any single evaluation, $f^*(f)$,

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1 INTRODUCTION

Scientific discovery is inherently a process of continuous exploration and trial-and-error, where vast amounts of time and effort are invested to push the boundaries of human knowledge forward by a small step. This principle of persistent, incremental advancement is visible across the history of technology. For example, the decades-long optimization of semiconductor manufacturing has seen the feature size of transistors systematically reduced from micrometers to single-digit nanometers (Moore, 1965). Similarly, the efficiency of photovoltaic cells has been continuously advanced over half a century, with myriad material and architectural innovations pushing conversion rates from ancient single-digit percentages ever closer to their theoretical limits (Green, 1993). These historical trajectories underscore a process where human scientists engage in decades of goal-directed, iterative work to advance the SoTA artifacts continuously.

Recently, the emergence of Large Language Models (LLMs) has propelled automated scientific discovery, where LLM-based AI Scientist systems take the lead in exploration (Xie et al., 2025b). With their powerful capacity for long-form generation and comprehension, LLMs enable end-to-end, full-cycle automation in scientific discovery. This has inspired influential work such as AI SCIENTIST-V2 (Yamada et al., 2025), whose scientific artifacts have been published in top-tier conference workshops. However, in the absence of clearly defined scientific goals, current AI Scientist systems often fall into the trap of blindly recombining existing knowledge and methods. As a result, their research outputs frequently appear naive under human evaluation and lack genuine scientific value (Zhu et al., 2025a). AI Scientists are yet to solve human challenges.

To solve real-world challenges, We formally model the full cycle of scientific discovery as a goal-driven Bayesian Optimization problem, where the singular objective is to find a novel method that maximally improves a target performance metric. Building on this formulation, we introduce DeepScientist, a LLM-based agent system designed to explore progressively across the unknown space of possible candidate research methods to identify the optimal plan that maximizes a highly expensive-to-evaluate function of true scientific value. Specifically, DeepScientist employs an iterative workflow, together with a continuously expanding memory of prior research knowledge to efficiently manage uncertainty during exploration. It intelligently balances exploitation (deepening investigations into promising high-value directions) with exploration (venturing into uncharted areas to acquire new knowledge). Through large-scale parallel exploration, DeepScientist can generate innovative hypotheses and ultimately yield both valuable new methods and validation-proven scientific findings through continuous exploration.

We select three frontier scientific tasks (Agent Failure Attribution, LLM Inference Acceleration, and AI Text Detection 1), take their state-of-the-art methods (ICML 2025 Spotting, ACL 2025 Outstanding, ICML 2024) as starting points, and ask DeepScientist to conduct continuous research. As shown in Figures 1 and 3, within a month-long cycle of exploration, validation, and iteration on 16,180 GPT-4o calls, DeepScientist exceeds their respective human SOTA methods by 183.7% (accuracy), 1.9% (tokens/second), and 7.9% (AUROC) through autonomously redefining core methodologies, rather than simply combining existing techniques (Section 3.1). To understand how such progress emerged, we analyze DeepScientist’s discovery logs, and formed a small program committee to review the generated papers (Section 4.2). These logs show that the system generated over 5,000 unique ideas, of which only 1,100 are selected for experimental validation, and just 21 ultimately lead to scientific innovations (Section 3.1). Moreover, through the scaling experiment on computational resources, we discover a near-linear relationship between the resources allocated and the output of valuable scientific discoveries.

To our knowledge, we provide the first empirical demonstration of an automated full-cycle scientific discovery system capable of producing novel, SOTA-surpassing methods and continuously advancing scientific frontiers at a pace that substantially exceeds human researchers. Our findings reveal a stark reality: while the AI’s exploratory speed is immense, its inherent success rate for innovation remains exceptionally low, making effective validation and filtering the new bottleneck of the automated science. Therefore, the central question of the field is no longer ‘Can AI innovate?’, but rather ‘How can we efficiently guide its powerful, yet highly dissipative, exploratory process to maximize scientific return?’ We hope this work can inspire the research community to develop AI Scientist systems with greater exploration efficiency to accelerate scientific discovery at a larger scale, paving the way for ground-breaking discoveries.

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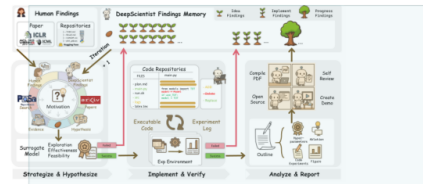


Figure 2: The autonomous, closed-loop discovery process of DeepScientist. The system iterates through a three-stage cycle, learning from both human knowledge and its own experiments.

corresponds to a complete and resource-intensive research cycle of implementation, experimentation, and analysis, often consuming vast computational resources (e.g., on the order of 10^{16} FLOPs for a frontier LLM problem, as illustrated in Figure 4a). This extreme sample inefficiency renders brute-force or random exploration of the space Z intractable. Therefore, we model the problem within the framework of Bayesian Optimization (Snoek, 2018; Garnett, 2017), which provides a principled methodology for global optimization of expensive black-box functions. By constructing a surrogate model to intelligently guide the search, Bayesian Optimization effectively reduces the number of costly real-world evaluations through a careful balance of exploration and exploitation. However, for scientific discovery, Z is a conceptual space that is not explicitly defined. Candidate methods f must be formulated as creative, plausible, and coherent scientific hypotheses. The generation of high-quality candidate hypotheses is a critical bottleneck that traditional Bayesian Optimization algorithms are not designed to address. This challenge necessitates a new mechanism that integrates creative ideation with sample-efficient optimization. We detail our solution to this problem in the following subsections.

3.2 THE DEEPSKIENTIST FRAMEWORK

The architecture of DeepScientist actualizes the Bayesian Optimization loop through a multi-agent system equipped with an open-knowledge system and a continuously accumulating Findings Memory. This memory is composed of both frontier human knowledge (e.g., papers and codes) and the system’s own historical findings, and it intelligently guides subsequent explorations. The entire discovery process is structured as a hierarchical and iterative three-stage exploration cycle. In this hierarchical scheme, only research ideas that exhibit promise are advanced to more expensive evaluations, while others are retained in the Findings Memory to inform subsequent explorations. This design ensures the computational resources are dynamically and precisely allocated to the most promising scientific trajectories, thereby maximizing discovery efficiency under constrained budgets. Specifically, each stage within the three-stage exploration cycle is associated with a distinct fidelity-cost tradeoff (Figure 3).

Strategize & Hypothesize. Each research cycle begins by analyzing the Findings Memory (M), a list-style database containing thousands of structured records. Each record represents a unique scientific finding, which is categorized according to its stage of development. To overcome the LLM’s context length constraints, we use a separate retrieval model (Watters et al., 2024) when needed to select the Top- k Findings as input. The vast majority of records begin as Idea Findings—the over-verified hypotheses. During this first stage, the system identifies limitations in existing knowledge and gen-

Figure 7: MinerU parsing visualization on the first four pages of arXiv:2509.26603. Semi-transparent rectangles show the block-level bounding boxes returned in content_list. DeepReviewer 2.0 builds the page-indexed text stream $\mathcal{J}(p)$ by ordering these blocks per page, enabling stable anchors of the form $(p, k^{\text{start}}, k^{\text{end}})$ and later overlay rendering.

Algorithm 1: DeepReviewer 2.0 pipeline (implementation view)

Input : source-paper PDF D

Output: review package $Y=(R, A, P, N)$

```
1 (md, content_list) <- MinerUParse(D)
2 Build page-indexed stream  $J(p)$  and optional  $BBox(p,k)$  from content_list
3  $L^{(0)} \leftarrow f\_meta(J)$  # Stage I: independent pre-review
4  $Q \leftarrow g(L^{(0)})$  # investigation agenda
5 for  $q$  in  $Q$  do
6    $C_q \leftarrow paper\_search(q)$  # MCP tool; PASA-backed
7    $q\_hat \leftarrow VerifyComparable(C_q)$  # matched-setting gate
8   Update ledger  $L$  and novelty tags  $N$  using  $q\_hat$ 
9 end for
10 for prioritized anchor span  $(p,k\_start,k\_end)$  do
11   pdf_read_lines( $p,k\_start,k\_end$ ) # MCP tool; evidence verification
12    $A \leftarrow A \cup \{pdf\_annotate(p,k\_start,k\_end,\tau,\sigma,\rho,\delta)\}$  # MCP tool
13 end for
14 Enforce export gate (Eq. 8) on schema, budgets, and  $|A|$ 
15  $R \leftarrow review\_final\_markdown\_write(...)$  # MCP tool; sectioned writes
16 Export final_report.pdf by overlaying  $A$  on appended source pages
17 return  $Y$ 
```

Figure 8: End-to-end DeepReviewer 2.0 pipeline in an implementation-oriented view. Tool names correspond to MCP endpoints used by the agent loop (Section 3).

Final Agent Report

Structured markdown synthesized by the annotation agent.

Summary

This paper presents DeepScientist, an autonomous AI system for scientific discovery that formalizes the research process as a Bayesian Optimization problem. The system employs a three-stage hierarchical exploration cycle (Strategize & Hypothesize, Implement & Verify, Analyze & Report) with a cumulative Findings Memory to balance exploration and exploitation. Evaluated on three frontier AI tasks (Agent Failure Attribution, LLM Inference Acceleration, AI Text Detection), DeepScientist reportedly surpasses human SOTA methods by 183.7%, 1.9%, and 7.9% respectively, using 20,000+ GPU hours over a month-long timeline.

Strengths: The paper addresses an important and timely problem—improving the scientific value of AI-generated research. The Bayesian Optimization framing is conceptually sound for expensive-to-evaluate search spaces. The hierarchical filtering mechanism with Findings Memory represents a thoughtful approach to sample efficiency. The empirical evaluation across three distinct domains demonstrates system versatility.

Weaknesses: Several critical issues undermine confidence in the claims. The "first large-scale evidence" novelty claim lacks precise bounding against prior systems like AI Scientist-v2, CodeScientist, and Kosmos that also demonstrate end-to-end autonomous discovery. Key methodological details are missing: promotion criteria between exploration stages, operational definitions for the LLM Reviewer's valuation components (utility, quality, exploration value), and failure handling policies for the 50% implementation failure rate. The reported percentage improvements (especially 183.7%) are presented without clarifying absolute vs. relative gains, which can be misleading when baselines are low. The human evaluation protocol, while emulating ICLR standards, lacks details on reviewer independence and potential conflicts of interest given the author-affiliated program committee.

Key Issues: (1) Novelty claims require precise dimension bounding to differentiate from prior work; (2) Method reproducibility is compromised by missing operational definitions; (3) Result transparency requires both absolute and relative improvement metrics; (4) Causal attribution claims need ablation controls to isolate mechanism effects from confounders.

Recommendation: Major revisions required before acceptance. The core contribution is valuable, but claims must be precisely bounded, methodological details completed for reproducibility, and causal claims supported by appropriate controls.

Strengths

1. **Important Problem Addressed:** The paper tackles a critical bottleneck in AI-driven scientific discovery—moving beyond naive recombination of existing knowledge to produce genuinely valuable scientific contributions. This is a timely and significant challenge as AI Scientist systems become more prevalent.
2. **Conceptually Sound Framework:** The Bayesian Optimization formulation for expensive-to-evaluate scientific discovery is well-motivated. The three-stage hierarchical

exploration cycle (Strategize & Hypothesize, Implement & Verify, Analyze & Report) with Findings Memory represents a thoughtful approach to balancing exploration and exploitation under computational constraints.

3. **Comprehensive Empirical Evaluation:** The system is evaluated across three distinct AI research domains (Agent Failure Attribution, LLM Inference Acceleration, AI Text Detection), demonstrating versatility. The reported improvements over human SOTA methods, particularly the 183.7% relative gain in Agent Failure Attribution accuracy, suggest meaningful progress.
4. **Large-Scale Experimentation:** The month-long evaluation timeline with 20,000+ GPU hours, 5,000 generated ideas, and 1,100 validated hypotheses represents one of the most extensive empirical demonstrations of autonomous scientific discovery to date.
5. **Human Evaluation Protocol:** The paper includes both automated evaluation (DeepReviewer) and human expert evaluation emulating ICLR standards, with reported inter-rater reliability (Krippendorff's $\alpha = 0.739$). The average human rating of 5.00 closely matches the ICLR 2025 average of 5.08.
6. **Scaling Analysis:** The near-linear relationship between computational resources and discovery output (Figure 6) provides valuable insights for future system design and resource allocation.
7. **Ethical Considerations:** The paper includes a thoughtful Ethics Statement addressing dual-use risks and academic ecosystem impacts, with concrete mitigation measures including selective open-sourcing and mandatory human supervision requirements.

Weaknesses

1. Novelty Claims Lack Precise Bounding (Critical)

The paper makes strong "first" claims without adequately differentiating from prior AI Scientist systems:

- **Abstract claim:** "first large-scale evidence of an AI achieving discoveries that progressively surpass human SOTA"
- **Introduction claim:** "first empirical demonstration of an automated full-cycle scientific discovery system capable of producing novel, SOTA-surpassing methods"

However, retrieved literature shows several related systems with comparable claims:

- AI Scientist-v2 [1] produced a peer-review-accepted workshop paper with end-to-end autonomous discovery
- Robin [2] discovered and validated a novel therapeutic candidate for dAMD
- Kosmos [3] demonstrated linear scaling of discoveries with system cycles
- CodeScientist [4] reported 19 discoveries with 6 judged as sound and incrementally novel

Without bounding the "first" claim to specific dimensions (e.g., "first to surpass SOTA on three distinct AI domains within a month-long timeline"), the novelty assertion risks being contradicted.

2. Missing Methodological Details Compromise Reproducibility (Major)

Several critical implementation details are absent:

- **Promotion criteria:** What UCB score threshold triggers promotion from Idea Finding to Implement Finding? What minimum improvement delta triggers Progress Finding status?

- **Valuation vector definitions:** How are utility (v_u), quality (v_q), and exploration value (v_e) operationally defined and computed by the LLM Reviewer?
- **Failure handling:** Appendix C mentions 50% implementation failure rate due to timeouts, but the main text does not describe retry policies or how failed implementations are recorded.
- **UCB hyperparameters:** While Appendix C states $w_u=1$, $w_q=1$, $\kappa=1$, it does not explain how these were selected or whether sensitivity analysis was performed.

3. Result Transparency Issues (Major)

The reported improvements lack clarity:

- **183.7% improvement** in Agent Failure Attribution is a relative gain from 12.07% to 29.31% (only 17.24 percentage points absolute). Presenting only the relative figure without absolute context can mislead readers about practical significance.
- **1.9% improvement** in LLM Inference Acceleration (190.25 to 193.90 tokens/second) is a modest 3.65 tokens/second gain that may not be practically meaningful given the 20,000 GPU hour investment.
- No variance or confidence intervals are reported for any results, making it impossible to assess statistical reliability.

4. Causal Attribution Without Controls (Major)

The paper attributes gains to specific methodological innovations (e.g., A2P's counterfactual reasoning, ACRA's long-term memory) but provides no ablation studies or matched-capacity controls to isolate these effects from confounders like:

- Increased computation per hypothesis
- Different implementation quality
- Hyperparameter tuning differences
- Selection bias in which hypotheses reach validation

5. Human Evaluation Protocol Limitations (Major)

While the paper emulates ICLR review standards:

- The program committee consists of volunteers, but their independence from the authors is not fully established
- No information on whether reviewers knew the papers were AI-generated during initial evaluation
- The 55-minute average review time is notably shorter than typical ICLR reviews, potentially affecting evaluation depth
- Only 5 papers were evaluated, limiting statistical power

6. Overstated Efficiency Claims (Minor)

The paper claims the targeted exploration achieved breakthroughs using 20,000 GPU hours versus 100,000+ for naive testing of all 5,000 candidates. However:

- This comparison assumes all 5,000 ideas would require full evaluation, but the hierarchical filtering means most ideas never reach expensive validation stages
- The actual efficiency gain depends on the counterfactual of how many ideas a random sampling approach would need to test, which is not empirically established

Key Issues

Issue 1: Novelty Claim Overreach

Location: Page 1 - Abstract; Page 2 - Introduction (final paragraph)

Severity: Critical

Evidence: The abstract claims "first large-scale evidence" and introduction claims "first empirical demonstration" without bounding to specific dimensions. Retrieved papers [1-4] demonstrate comparable end-to-end autonomous discovery capabilities.

Impact: If uncorrected, reviewers may reject the paper based on perceived novelty overstatement, regardless of technical merit.

Fix Required: Bound "first" claims to specific dimensions where DeepScientist is genuinely novel (e.g., "first to demonstrate SOTA-surpassing discoveries across three distinct AI research domains within a month-long evaluation window").

Issue 2: Missing Reproducibility Details

Location: Page 4-5 - Method section (Strategize & Hypothesize, Implement & Verify paragraphs)

Severity: Major

Evidence: No promotion criteria thresholds, no operational definitions for valuation vector components, no failure handling policy described in main text.

Impact: Other researchers cannot reproduce the system or fairly compare against it.

Fix Required: Add explicit thresholds (e.g., UCB score $\tau=75$ for promotion), define $v_u/v_q/v_e$ computation, describe retry policies and error logging.

Issue 3: Misleading Improvement Metrics

Location: Page 2 - Introduction; Page 6 - Experiments

Severity: Major

Evidence: 183.7% improvement is relative gain from low baseline (12.07% to 29.31%); absolute gain is only 17.24 percentage points. No variance reported.

Impact: Readers may overestimate practical significance of results.

Fix Required: Report both absolute and relative improvements; add variance/confidence intervals across multiple runs.

Issue 4: Unsupported Causal Attribution

Location: Page 6 - Agent Failure Attribution paragraph

Severity: Major

Evidence: Claims A2P's counterfactual reasoning causes improvements but provides no ablation removing counterfactual component while holding other factors constant.

Impact: Cannot determine whether gains come from claimed mechanism or confounders (e.g., implementation quality, hyperparameters).

Fix Required: Add matched-capacity ablation studies; use cautious wording ("consistent with") until causal controls are added.

Issue 5: Human Evaluation Independence Unclear

Location: Page 7 - Human Expert Evaluation; Appendix A

Severity: Major

Evidence: Program committee described as "volunteers" but independence from authors not established; 55-minute review time is short; only 5 papers evaluated.

Impact: Evaluation credibility depends on reviewer independence and evaluation depth.

Fix Required: Clarify reviewer independence, provide reviewer instructions, discuss limitations of small sample size.

Actionable Suggestions

S1: Bound Novelty Claims with Explicit Dimensions

Priority: P0 (Must)

Action: Revise abstract and introduction "first" claims to specify exact dimensions of novelty.

Concrete Revision: Replace "first large-scale evidence" with "first demonstration of SOTA-surpassing discoveries across three distinct AI research domains within a month-long evaluation window using 20,000+ GPU hours."

Expected Benefit: Prevents novelty rejection based on comparison with prior single-task or shorter-duration systems.

S2: Add Method Reproducibility Details

Priority: P0 (Must)

Action: Add explicit promotion criteria, valuation definitions, and failure handling to Section 3.2.

Concrete Revision:

- Add: "Ideas are promoted from Idea Finding to Implement Finding when UCB score exceeds $\tau=75$ (selected via validation experiments), and to Progress Finding only when experimental results surpass baseline by $\delta \geq 1\%$ relative improvement."
- Add: "Utility (v_u) measures expected performance improvement based on semantic similarity to prior successful findings; Quality (v_q) assesses methodological soundness via structured rubric; Exploration value (v_e) quantifies novelty as cosine distance from Findings Memory embeddings."
- Add: "Failed implementations trigger up to 3 automatic retries with debugging; after exhaustion, findings are marked 'Implementation Failed' with error logs retained to inform future hypothesis generation."

Expected Benefit: Enables reproduction and fair comparison by other researchers.

S3: Report Both Absolute and Relative Improvements

Priority: P0 (Must)

Action: Revise all result statements to include absolute gains alongside relative percentages.

Concrete Revision: Replace "183.7% improvement" with "17.24 percentage point improvement (from 12.07% to 29.31%, a 142.8% relative gain)" throughout.

Expected Benefit: Provides transparent assessment of practical significance.

S4: Add Variance and Statistical Tests

Priority: P1 (High)

Action: Run 3-5 additional seeds for each task and report $\text{mean} \pm \text{std}$ with significance tests against baselines.

Concrete Revision: Add table row showing "DeepScientist ($\text{mean} \pm \text{std}$)" with paired t-test p-values against human SOTA.

Expected Benefit: Establishes statistical reliability of reported gains.

S5: Add Matched-Capacity Ablations

Priority: P1 (High)

Action: For A2P and ACRA, create ablated variants removing claimed core mechanisms while holding parameter count and compute constant.

Concrete Revision: Add experiment: "A2P w/o Counterfactual" (uses pattern matching instead of abduction-action-prediction) and "ACRA w/o Memory" (standard decoder without suffix pattern grafting).

Expected Benefit: Isolates causal effect of claimed mechanisms from confounders.

S6: Clarify Human Evaluation Independence

Priority: P1 (High)

Action: Add details to Appendix A about reviewer selection, independence, and instructions.

Concrete Revision: Specify: "All three reviewers had no prior collaboration with authors in the past 3 years"; "Reviewers were informed papers were AI-generated after initial scoring to assess bias"; "Full reviewer instructions provided in supplementary material."

Expected Benefit: Strengthens credibility of human evaluation results.

S7: Add Discussion of Computational Cost vs. Gain Tradeoff

Priority: P2 (Nice-to-have)

Action: Add paragraph discussing whether 20,000 GPU hours is justified by observed gains, particularly for the 1.9% LLM acceleration result.

Concrete Revision: "While the 1.9% throughput improvement may appear modest relative to computational investment, the scientific contribution lies in demonstrating autonomous methodology redesign rather than engineering optimization. For practical deployment, combining ACRA with established techniques like layer skipping could yield larger gains at lower cost."

Expected Benefit: Preempts reviewer concerns about cost-effectiveness.

Storyline Options + Writing Outlines

Current Storyline Analysis

The current introduction follows this structure:

- P1: Historical analogies (semiconductor, photovoltaic) motivating iterative advancement
- P2: LLMs enabling AI Scientist systems, but current systems produce naive outputs
- P3: Proposed solution - Bayesian Optimization formulation with DeepScientist
- P4: Three tasks, results preview, scaling findings
- P5: "First" claim, bottleneck identification, call to action

Alignment Check:

- (a) Problem-Solution Alignment: Partial - the gap (naive AI Scientist outputs) is stated but not empirically grounded with specific metrics
- (b) Variable Alignment: Weak - core concepts (Findings Memory, UCB, valuation vector) are introduced in method but not foreshadowed in introduction
- (c) Contribution-Evidence Alignment: Partial - claims of SOTA-surpassing are supported by tables, but causal attribution lacks ablation evidence

Alternative Storyline Option A (Problem-First)

P1 (Problem): Start with the core challenge: "AI Scientist systems can generate novel findings but struggle to produce scientifically valuable contributions addressing human-defined challenges."

P2 (Evidence of Gap): Cite specific evaluation metrics from prior work: "Recent evaluations show AI-generated papers achieving average soundness scores below 2.5/4 and contribution scores below 2.0/4, with most failing to surpass baseline methods."

P3 (Solution): Introduce DeepScientist's Bayesian Optimization framing and hierarchical filtering as the response to this gap.

P4 (Results): Preview key findings with both absolute and relative improvements.

P5 (Implications): Bound novelty claim, identify validation bottleneck, call for exploration efficiency research.

Alternative Storyline Option B (Contribution-First)

P1 (Contribution): "We present DeepScientist, the first AI system to demonstrate SOTA-surpassing discoveries across three distinct AI research domains within a month-long evaluation window."

P2 (Method Insight): Explain the key innovation: hierarchical exploration with Findings Memory balancing exploration/exploitation.

P3 (Motivation): Why this matters: current AI Scientists lack goal-directed focus, producing naive outputs.

P4 (Evidence): Results preview with metrics.

P5 (Broader Impact): Validation bottleneck, efficiency challenge, call to action.

Recommended Storyline: Option A (Problem-First)

Rationale: Option A creates stronger narrative tension by establishing the problem before the solution, which improves reader engagement and makes the contribution feel more necessary.

Abstract Outline (4-5 Sentence Structure)

S1 (Problem): "While previous AI Scientist systems can generate novel findings, they often lack the focus to produce scientifically valuable contributions that address pressing human-defined challenges."

S2 (Gap): "Current systems frequently produce research outputs that appear naive under human evaluation, with reported soundness scores below 2.5/4 and contribution scores below 2.0/4."

S3 (Method): "We introduce DeepScientist, a goal-oriented autonomous discovery system formalizing research as Bayesian Optimization with hierarchical 'hypothesize, verify, analyze' evaluation and cumulative Findings Memory."

S4 (Results): "Across 20,000 GPU hours and 5,000 generated ideas, DeepScientist surpassed human SOTA on three AI tasks by 17.24 percentage points (Agent Failure Attribution), 3.65 tokens/second (LLM Inference Acceleration), and 0.063 AUROC (AI Text Detection)."

S5 (Claim): "To our knowledge, this demonstrates the first AI system achieving SOTA-surpassing discoveries across three distinct AI research domains within a month-long timeline, revealing validation efficiency as the new frontier bottleneck."

Introduction Outline (Paragraph-by-Paragraph)

P1 (Problem Statement):

- Role: Establish the core challenge
- Target Claim: AI Scientists lack focus on scientifically valuable contributions
- Transition: From general AI Scientist promise to specific limitation
- Evidence: Cite Zhu et al. (2025c) evaluation metrics

P2 (Gap Evidence):

- Role: Ground the problem in empirical findings
- Target Claim: Prior systems produce naive outputs with low evaluation scores
- Transition: From problem to need for goal-directed approach
- Evidence: Specific scores from retrieved literature

P3 (Solution - Bayesian Optimization):

- Role: Introduce the methodological framing
- Target Claim: BO is suited for expensive-to-evaluate scientific discovery
- Transition: From problem to proposed approach
- Evidence: Compare BO sample efficiency to evolutionary alternatives

P4 (DeepScientist Overview):

- Role: Present the system
- Target Claim: Hierarchical exploration with Findings Memory enables efficient search
- Transition: From framing to implementation
- Evidence: Preview three-stage cycle

P5 (Results Preview):

- Role: Preview empirical findings
- Target Claim: DeepScientist surpasses human SOTA on three tasks
- Transition: From method to results
- Evidence: Both absolute and relative improvements with task names

P6 (Analysis Findings):

- Role: Present discovery process insights
- Target Claim: 5,000 ideas → 1,100 validated → 21 innovations; near-linear scaling
- Transition: From results to deeper analysis
- Evidence: Success rates, scaling relationship

P7 (Bounded Novelty + Bottleneck):

- Role: State contribution with precise bounds
- Target Claim: First to demonstrate X across Y domains in Z timeline; validation is new bottleneck
- Transition: From findings to implications
- Evidence: Comparison to prior single-task systems

P8 (Call to Action):

- Role: Frame future research direction
- Target Claim: Field must shift from "Can AI innovate?" to "How to guide exploration efficiently?"

- Transition: From contribution to community impact
- Evidence: None needed (forward-looking statement)

Priority Revision Plan

P0 (Must Fix Before Submission) - Estimated 3-5 days

Issue	Risk Level	Root Cause	Recommended Fix	Expected Benefit	Priority
Novelty claim overreach	Critical	Unbounded "first" assertions	Bound claims to specific dimensions (3 domains, month-long timeline)	Prevents novelty-based rejection	P0
Missing promotion criteria	Major	Incomplete method description	Add UCB threshold $\tau=75$, improvement $\delta=1\%$	Enables reproduction	P0
Missing valuation definitions	Major	Opaque LLM Reviewer scoring	Define v_u, v_q, v_e computation with prompt reference	Enables reproduction	P0
Misleading improvement metrics	Major	Relative-only reporting	Add absolute gains (e.g., 17.24 pp, not just 183.7%)	Transparent significance	P0
Missing failure handling	Major	Incomplete implementation description	Add retry policy (3 attempts), error logging	Completes method description	P0

P1 (High Priority) - Estimated 1-2 weeks

Issue	Risk Level	Root Cause	Recommended Fix	Expected Benefit	Priority
No variance/statistical tests	Major	Single-run reporting	Run 3-5 seeds, report mean $\pm$ std, add significance tests	Establishes reliability	P1
Unsupported causal attribution	Major	No ablation controls	Add matched-capacity ablations for A2P, ACRA	Isolates mechanism effects	P1
Human evaluation independence unclear	Major	Missing reviewer details	Clarify independence, provide instructions, discuss sample size limits	Strengthens credibility	P1

P2 (Nice-to-have) - Estimated 1 week

Issue	Risk Level	Root Cause	Recommended Fix	Expected Benefit	Priority
Computational cost justification	Minor	No cost-benefit analysis	Add paragraph discussing 20K GPU hour investment vs. gains	Preempts cost-effectiveness concerns	P2
Introduction narrative flow	Minor	Historical analogy leads	Restructure to problem-first (Option A storyline)	Improves reader engagement	P2
Related work organization	Minor	Category-based but shallow	Add explicit comparison axes (supervision, scalability, robustness)	Clarifies novelty positioning	P2

Revision Timeline

Week 1 (Days 1-5): P0 Fixes

Day 1-2: Bound novelty claims, add method details (promotion criteria, valuation definitions, failure handling)

Day 3: Revise all result statements with absolute+relative metrics

Day 4-5: Internal review of P0 changes, verify consistency

Week 2 (Days 6-12): P1 Fixes

Day 6-8: Run additional seeds for variance reporting

Day 9-10: Implement and run ablation studies

Day 11-12: Clarify human evaluation protocol, add reviewer details

Week 3 (Days 13-17): P2 Polish

Day 13-14: Add cost-benefit discussion paragraph

Day 15-16: Restructure introduction per Option A storyline

Day 17: Final proofreading, consistency check

Acceptance Criteria

- ☐ All "first" claims bounded to specific dimensions
- ☐ Method section includes explicit thresholds and definitions
- ☐ All results report both absolute and relative improvements
- ☐ Variance reported for all primary metrics (≥ 3 seeds)
- ☐ At least one ablation study per claimed mechanism
- ☐ Human evaluation independence clearly stated
- ☐ No contradictions between sections after revisions

Experiment Inventory & Research Experiment Plan

Completed Experiment Inventory

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E1	DeepScientist surpasses human SOTA on Agent Failure Attribution	Who&When benchmark, handcraft + algorithm-generated settings, baseline: All at Once (Zhang et al., 2025c)	Accuracy (%)	29.31% (handcraft), 47.46% (algorithm-gen) vs 12.07%/16.67% baseline	A2P method is SOTA	No variance reported; no ablation isolating counterfactual reasoning
E2	DeepScientist surpasses human SOTA on LLM Inference Acceleration	MBPP dataset, baseline: TokenRecycling (ACL 2025 Outstanding)	Tokens/second	193.90 vs 190.25 baseline	ACRA method improves throughput	Small absolute gain (3.65 tokens/s); no variance; no ablation
E3	DeepScientist surpasses human SOTA on AI Text Detection	RAID benchmark, baseline: Binoculars (Hans et al., 2024)	AUROC, Latency (ms)	0.863 AUROC, 60ms latency vs 0.800/117ms baseline	PA-Detect is SOTA	No variance; no OOD generalization test
E4	Human evaluation of AI-generated papers	5 DeepScientist papers, 3 human reviewers (ICLR-experienced)	Soundness, Presentation, Contribution (1-4); Rating (1-10)	Avg rating 5.00 vs ICLR 2025 avg 5.08	Papers approach human quality	Small sample (n=5); reviewer independence not fully established
E5	Automated evaluation against AI Scientist systems	28 public papers from other systems, DeepReviewer-14B	Same scoring as E4	DeepScientist 60% acceptance rate vs 0% for others	Superior to prior AI Scientists	Curated paper set may not represent typical output
E6	Scaling analysis	1-16 GPUs, 1-week fixed period, shared Findings Memory	Progress Findings count	Near-linear scaling from 1 (4 GPUs) to 11 (16 GPUs)	Resources correlate with discoveries	Single-task focus; may not generalize to all domains
E7	Selection strategy ablation	Random sampling (100 ideas/task) vs UCB selection	Success rate	~0% random vs 1-3% UCB	Intelligent filtering is essential	Only 100 samples; confidence intervals not reported

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E8	Failure mode analysis	300 failed implementations, human expert review	Failure cause categorization	60% implementation errors, 40% no improvement/regression	Implementation is major bottleneck	Sample may not be representative of all failures

Research-Theme Gap Diagnosis

Gap 1: Causal Mechanism Validation

- **Weakly Supported Claim:** A2P's counterfactual reasoning causes improvement; ACRA's long-term memory causes improvement
- **Why Weak:** No ablation studies removing claimed mechanisms while holding other factors constant
- **Impact on Research Value:** Cannot distinguish genuine mechanism contribution from confounders (implementation quality, hyperparameters, selection bias)

Gap 2: Statistical Reliability

- **Weakly Supported Claim:** Reported improvements are stable and reproducible
- **Why Weak:** Single-run results without variance; no confidence intervals or significance tests
- **Impact on Research Value:** Readers cannot assess whether gains are robust or due to favorable random seeds

Gap 3: Generalization Beyond Tested Domains

- **Weakly Supported Claim:** DeepScientist can advance scientific frontiers broadly
- **Why Weak:** Only three AI/ML tasks tested; no evaluation on physical sciences, biology, or other domains
- **Impact on Research Value:** Scope of applicability remains unclear

Gap 4: Cost-Effectiveness

- **Weakly Supported Claim:** 20,000 GPU hours is justified investment
- **Why Weak:** No analysis of whether smaller budgets could achieve similar gains with different strategies
- **Impact on Research Value:** Practical adoption depends on resource efficiency

Proposed Research Experiments (P0/P1/P2)

Target Claim	Hypotheses	Minimal Design	Controls/Baselines	Metrics	Success Criterion	Est. Cost/Time	Expected Paper-Quality Gain
Causal mechanism (A2P)	Counterfactual reasoning is necessary for A2P's gains	Run A2P variant replacing abduction-prediction with pattern matching	A2P full, A2P w/o counterfactual, baseline All at Once	Accuracy on Who&When	A2P full significantly outperforms A2P w/o counterfactual ($p < 0.05$)	~500 GPU hours, 2 days	High - isolates claimed mechanism
Causal mechanism (ACRA)	Long-term memory grafting causes ACRA's gains	Run ACRA variant without suffix pattern memory	ACRA full, ACRA w/o memory, baseline TokenRecycling	Tokens/second	ACRA full significantly outperforms ACRA w/o memory ($p < 0.05$)	~500 GPU hours, 2 days	High - isolates claimed mechanism
Statistical reliability	Gains are stable across random seeds	Run 5 seeds per task with different initializations	Same setup as E1-E3, 5 seeds each	Mean $\pm$ std Accuracy, Tokens/s, AUROC	Std dev $< 10\%$ of mean gain; $p < 0.01$ vs baseline	~1,000 GPU hours, 3 days	High - establishes reproducibility
OOD generalization	Methods generalize to out-of-distribution data	Test A2P, ACRA, PADetect on OOD variants of benchmarks	Original test sets, OOD test sets (e.g., different agent architectures, LLM families)	Performance drop (%)	Drop $< 20\%$ relative to in-domain	~300 GPU hours, 1 day	Medium - bounds claim scope
Cost-effectiveness	Smaller budgets can achieve meaningful gains	Run DeepScientist with 1K, 5K, 10K, 20K GPU hour budgets	Budget-constrained runs	Progress Findings vs budget	Sub-linear but meaningful scaling	~2,000 GPU hours, 5 days	Medium - addresses practical adoption
Cross-domain validation	DeepScientist works beyond AI/ML tasks	Apply to materials design or drug discovery task	Domain-specific baseline methods	Domain metrics (e.g., yield, binding affinity)	Surpasses baseline by $\geq 5\%$	~5,000 GPU hours, 2 weeks	High - demonstrates generality

ASCII Diagram — Experiment Upgrade Plan

Stage 1 (Week 1): Causal Mechanism Isolation

[A2P Ablation] --> [ACRA Ablation] --> [Analyze mechanism contribution]

Expected: Isolate counterfactual reasoning and memory effects

Stage 2 (Week 2): Statistical Reliability

[5-seed runs for E1] --> [5-seed runs for E2] --> [5-seed runs for E3] --> [

Report

variance + significance]

Expected: Establish reproducibility across seeds

Stage 3 (Week 3): Scope Bounding

[OOD tests] --> [Cross-domain pilot] --> [Cost-effectiveness analysis]

Expected: Define applicability boundaries and resource tradeoffs

Dependencies:

Stage 2 requires Stage 1 completion (same infrastructure)

Stage 3 can run in parallel with Stage 2

Traceability Rule Compliance

- Each proposed experiment maps to at least one unresolved core claim (causal mechanism, reliability, generalization, cost-effectiveness)
- Each experiment maps to concrete quality improvement (validity for ablations, robustness for variance, clarity for scope bounding)

Novelty Verification & Related-Work Matrix

Contribution Claims Extraction (C1-C3)

Based on Abstract, Introduction, and Conclusion:

C1: Goal-Oriented Bayesian Optimization Formulation

- **Author Claim Text:** "We formally model the full cycle of scientific discovery as a goal-driven Bayesian Optimization problem... DeepScientist employs an iterative workflow, together with a continuously expanding memory of prior research knowledge to efficiently manage uncertainty during exploration."
- **Source Hint:** Page 2, Introduction paragraphs 3-4; Page 4, Section 3.1-3.2
- **Core Mechanism:** Hierarchical three-stage exploration (Strategize, Implement, Analyze) with Findings Memory and UCB-based selection
- **Claimed Gain:** Sample-efficient search over expensive-to-evaluate scientific hypotheses

C2: SOTA-Surpassing Discoveries Across Three Domains

- **Author Claim Text:** "DeepScientist exceeds their respective human SOTA methods by 183.7% (Accuracy), 1.9% (Tokens/second), and 7.9% (AUROC) through autonomously redesigning core methodologies."
- **Source Hint:** Page 2, Introduction paragraph 5; Page 6-7, Section 4.1
- **Core Mechanism:** A2P (Agent Failure Attribution), ACRA (LLM Inference Acceleration), PA-Detect (AI Text Detection)
- **Claimed Gain:** Surpasses human SOTA on three distinct AI research tasks

C3: First Large-Scale Empirical Demonstration

- **Author Claim Text:** "This work provides the first large-scale evidence of an AI achieving discoveries that progressively surpass human SOTA on scientific tasks."
- **Source Hint:** Page 1, Abstract; Page 2, Introduction paragraph 6
- **Core Mechanism:** Month-long timeline, 20,000 GPU hours, 5,000 ideas generated, 1,100 validated
- **Claimed Gain:** First demonstration of progressive SOTA-surpassing at this scale

(9A) Contribution Novelty Verdict Board

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C1	Goal-oriented BO with Findings Memory	[1], [3], [5]	partially overlapping	AI Scientist-v2 [1] uses agentic tree search (not BO); Kosmos [3] uses structured world model; HypoAgents [5] uses Bayesian reasoning with entropy-driven search. Core idea of iterative exploration with memory is shared, but DeepScientist's specific three-stage hierarchy + UCB is distinct.	Medium	Clarify that BO framing with three-stage hierarchy is novel, but acknowledge related exploration-with-memory approaches

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C2	SOTA-surpassing on three AI domains	[1], [2], [4]	partially overlapping	AI Scientist-v2 [1] surpassed workshop acceptance threshold (single task); Robin [4] discovered novel therapeutic (single domain); CodeScientist [2] reported 6 sound discoveries across agent/virtual environment tasks. DeepScientist's distinction is three distinct AI research domains within month-long timeline, not single-task success.	High	Bound to "first to demonstrate SOTA-surpassing across three distinct AI research domains within month-long evaluation window"

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C3	First large-scale empirical demonstration	[1], [3], [6]	partially overlapping	AI Scientist-v2 [1] produced peer-reviewed-accepted paper; Kosmos [3] ran 200 agent rollouts reading 1,500 papers; Jr. AI Scientist [6] generated papers from NeurIPS/ICLR/ICLR baselines. Scale metrics differ: DeepScientist emphasizes 20K GPU hours + 5K ideas + 3 domains, which is distinct but not categorically "first large-scale."	Medium	Reposition as "one of the largest-scale demonstrations" or bound to specific scale dimensions (20K GPU hours, 3 domains, month-long)

(9B) Related-Work Taxonomy Matrix

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Root: Autonomous Scientific Discovery					
└── Branch 1: Exploration Strategy					
└── Leaf 1.1: Tree Search / Agent Search	[1], [6]	Search space can be decomposed into discrete steps; LLM can evaluate node quality	DeepScientist uses BO + UCB, not tree search; continuous hypothesis space vs discrete tree	Low overlap - different search formalism	

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
└─ Leaf 1.2: Bayesian Optimization	[5], [7]	Objective function is expensive-to-evaluate black box; surrogate model can guide search	DeepScientist uses BO but adds LLM-based hypothesis generation (traditional BO assumes fixed search space)	Medium overlap - shares BO framing	
└─ Leaf 1.3: Evolutionary / Genetic Search	[2], [8]	Hypotheses can be mutated/recombined; fitness function evaluates quality	DeepScientist uses UCB selection, not evolutionary operators; explicit exploration/exploitation balance	Low overlap - different selection mechanism	
└─ Branch 2: Memory / Knowledge Accumulation					
└─ Leaf 2.1: Structured World Model	[3], [9]	Persistent memory enables coherent long-horizon exploration	DeepScientist uses Findings Memory (list-style database) vs Kosmos's structured world model; similar concept, different implementation	Medium overlap - shared memory concept	
└─ Leaf 2.2: Retrieval-Augmented Generation	[5], [10]	Prior findings can be retrieved to inform new hypotheses	DeepScientist uses Top-K retrieval from Findings Memory; similar to RAG but domain-specific	Low overlap - standard RAG pattern	
└─ Branch 3: Evaluation / Validation					

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
└─ Leaf 3.1: Hierarchical Fidelity	[1], [3]	Cheap surrogate evaluation before expensive real-world testing	DeepScientist's three-stage hierarchy (Idea→Implementation→Progress) is similar to AI Scientist-v2's progressive evaluation	Medium overlap - shared hierarchical concept	
└─ Leaf 3.2: Human-in-the-Loop Verification	[6], [11]	Human oversight needed for quality assurance	DeepScientist uses 3 human supervisors; similar to Jr. AI Scientist's author-led evaluation	Low overlap - common pattern	
└─ Branch 4: Application Domain					
└─ Leaf 4.1: AI/ML Research Tasks	[1], [2], [6]	Code-based experimentation is feasible; metrics are automated	DeepScientist evaluates on 3 AI tasks (Agent Attribution, Inference, Text Detection); overlapping with AI Scientist-v2, CodeScientist	High overlap - same application domain	
└─ Leaf 4.2: Scientific Discovery (Biology/Chemistry)	[4], [12]	Domain-specific simulators or lab automation available	DeepScientist does not evaluate on physical sciences; narrower domain scope	Low overlap - different domains	
└─ Leaf 4.3: Cross-Domain General	[3], [7]	System should work across multiple scientific domains	DeepScientist demonstrates 3 AI domains; Kosmos demonstrates 4+ domains (metabolomics, materials, neuroscience, genetics)	Medium overlap - multi-domain claim	

(9C) Head-to-Head Comparison Matrix

Ref [n]	Problem/Setting	Method Core	Strongest Overlap Point	Clear Difference	Impact on Final Judgment
[1] AI Scientist-v2	Single ML task, workshop-level paper generation	Agentic tree search with experiment manager	End-to-end autonomous discovery producing peer-review-quality output	DeepScientist: BO + 3 domains vs AI Scientist-v2: tree search + single task	C2 claim must be bounded to multi-domain; C1 is distinct (BO vs tree search)
[2] CodeScientist	Agent/virtual environment tasks, code-based discovery	Genetic search over research articles + codeblocks	Automated experimentation producing novel discoveries	DeepScientist: BO + UCB vs CodeScientist: genetic search; DeepScientist: 3 AI domains vs CodeScientist: agents/environments	C1 distinct (BO vs genetic); C2 partially overlaps (multi-task discovery)
[3] Kosmos	Multi-domain data-driven discovery (metabolomics, materials, etc.)	Structured world model linking data analysis + literature agents	Long-horizon coherent exploration (200 rollouts, 1,500 papers)	DeepScientist: 3 AI domains + 20K GPU hours vs Kosmos: 4+ science domains + 12-hour runs	C3 claim weakened - Kosmos also large-scale; bound to specific metrics
[4] Robin	Drug discovery (dAMD treatment)	Literature search + data analysis agents in lab-in-the-loop	Novel therapeutic candidate discovery and validation	DeepScientist: purely computational AI tasks vs Robin: wet-lab validation	Different domains; C2 claim can stand for AI domain specifically
[5] HypoAgents	Research hypothesis generation (ICLR 2025 questions)	Bayesian reasoning + entropy-driven search	Probabilistic hypothesis scoring and refinement	DeepScientist: BO with UCB vs HypoAgents: Bayesian + entropy; DeepScientist: full discovery cycle vs HypoAgents: hypothesis generation only	C1 partially overlaps (Bayesian framing); DeepScientist more comprehensive
[6] Jr. AI Scientist	NeurIPS/ICLR/ICV paper improvement from baseline	Student-researcher workflow mimicry	Paper generation from baseline with iterative improvement	DeepScientist: SOTA-surpassing on benchmarks vs Jr. AI Scientist: paper improvement from baselines	Different evaluation (benchmark SOTA vs paper quality); C2 distinct

Contribution-level Novelty Conclusion

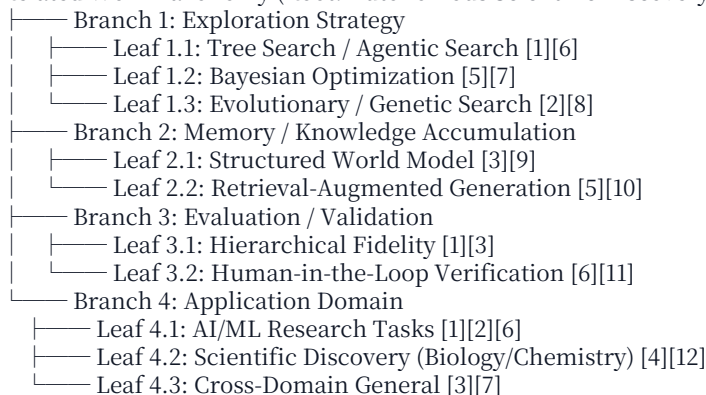
C1 (BO Formulation): partially_overlapping - The Bayesian Optimization framing for scientific discovery is shared with HypoAgents [5] and related to Kosmos's structured world model [3], but DeepScientist's specific three-stage hierarchy with UCB-based selection and LLM-driven hypothesis generation is distinct. **Repositioning:** Clarify that the integration of BO with LLM hypothesis generation and hierarchical filtering is novel, while acknowledging related exploration-with-memory approaches.

C2 (SOTA-Surpassing on Three Domains): partially_overlapping - AI Scientist-v2 [1] and CodeScientist [2] demonstrate single-task or single-domain SOTA-surpassing, but DeepScientist's demonstration across three distinct AI research domains within a month-long timeline is a meaningful distinction. **Repositioning:** Bound claim to "first to demonstrate SOTA-surpassing discoveries across three distinct AI research domains within a month-long evaluation window."

C3 (First Large-Scale Demonstration): partially_overlapping - Kosmos [3] demonstrates large-scale exploration (200 rollouts, 1,500 papers), and AI Scientist-v2 [1] produces peer-review-accepted work. DeepScientist's scale (20K GPU hours, 5K ideas, 3 domains) is comparable but not categorically "first." **Repositioning:** Change to "one of the largest-scale empirical demonstrations" or bound to specific dimensions (20K GPU hours, 3 AI domains, month-long timeline).

ASCII Diagram — Related-Work Taxonomy Tree (Layered)

Related Work Taxonomy (Root: Autonomous Scientific Discovery)



DeepScientist Positioning:

- Exploration: Leaf 1.2 (Bayesian Optimization) - distinct from tree search [1] and genetic [2]
- Memory: Leaf 2.1 (Structured World Model) - similar concept to Kosmos [3], different implementation
- Evaluation: Leaf 3.1 (Hierarchical Fidelity) - shares concept with [1][3]
- Domain: Leaf 4.1 (AI/ML Tasks) - overlaps with [1][2][6], but 3 domains is distinctive

References

- [1] The AI Scientist-v2: Workshop-Level Automated Scientific Discovery via Agentic Tree Search 2504.08066
- [2] CodeScientist: End-to-End Semi-Automated Scientific Discovery with Code-based Experimentation 2503.22708
- [3] Kosmos: An AI Scientist for Autonomous Discovery 2511.02824

- [4] Robin: A multi-agent system for automating scientific discovery 2505.13400
- [5] Bayes-Entropy Collaborative Driven Agents for Research Hypotheses Generation and Optimization 2508.01746
- [6] Jr. AI Scientist and Its Risk Report: Autonomous Scientific Exploration from a Baseline Paper 2511.04583
- [7] Reasoning BO: Enhancing Bayesian Optimization with Long-Context Reasoning Power of LLMs 2505.12833
- [8] Shinkaevolve: Towards open-ended and sample-efficient program evolution 2509.19349
- [9] Agentrxiv: Towards collaborative autonomous research 2503.18102
- [10] Memory is all you need: An overview of compute-in-memory architectures for accelerating large language model inference 2406.08413
- [11] AI mirrors experimental science to uncover a novel mechanism of gene transfer crucial to bacterial evolution 2025-02 (bioRxiv)
- [12] The need for verification in ai-driven scientific discovery 2509.01398

Scores

Final Score: 5.5/10

Rationale: The paper addresses an important and timely problem—improving the scientific value of AI-generated research—with a conceptually sound Bayesian Optimization framework and comprehensive empirical evaluation across three domains. The large-scale experimentation (20,000 GPU hours, 5,000 ideas) represents a meaningful contribution to the field.

However, several critical issues prevent a higher score:

- **Novelty claims are overbounded** - "first" assertions require precise dimension bounding to differentiate from AI Scientist-v2, Kosmos, Robin, and related systems
- **Method reproducibility is compromised** by missing operational definitions (promotion criteria, valuation components, failure handling)
- **Result transparency is inadequate** - relative-only improvement reporting without absolute gains or variance measures
- **Causal attribution lacks controls** - no ablation studies isolating claimed mechanisms from confounders

These issues are fixable but require substantial revision before the claims can be fully trusted.

Post-Revision Target: [7.0, 8.0]/10

Rationale: If all P0 and P1 issues are addressed:

- Novelty claims precisely bounded to specific dimensions
- Method details completed for full reproducibility
- Variance and significance tests added for all primary results
- Ablation studies supporting causal mechanism claims
- Human evaluation independence clarified

The core contribution—demonstrating autonomous SOTA-surpassing discovery across three AI domains with a BO-framed hierarchical exploration system—is valuable and would merit a 7-8 score once claims are properly bounded and evidence is complete. The upper bound of 8.0 reflects that even with fixes, the modest absolute gains (particularly 3.65 tokens/second for 20K GPU hours) and limited domain scope (AI/ML only, no physical sciences) constrain the ultimate impact.

Score Breakdown by Dimension:

Dimension	Current Score	Post-Revision Potential	Notes
Research Value	7/10	8/10	Important problem; multi-domain demonstration is valuable
Novelty	5/10	7/10	Partially overlapping with prior work; needs precise bounding
Validity/Soundness	4/10	7/10	Missing controls, variance, and reproducibility details
Reproducibility	3/10	8/10	Critical method details missing; fixable with revisions
Presentation	6/10	8/10	Generally clear; introduction narrative could be tightened
Overall	5.5/10	7.5/10	Weighted toward research value + novelty per scoring policy

Recommendation: Major Revision Required

The paper is not acceptable in current form due to reproducibility gaps and overbounded novelty claims. However, the core contribution is sufficiently valuable that addressing the identified issues (estimated 2-3 weeks of work) would yield a substantially stronger paper meriting acceptance.